Date 3th Aug 2023
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Digital Logic And Computer Organization

Logic Gate

Logic gate is an electronic circuit which can take one or more inputs but output will be one.

Note :-

① Logic gate are made up of diodes & transistors.
② A logic gate is used to allow and denied a digital signal.

Types :-

1. AND Gate
2. OR Gate
3. NOT Gate
4. NAND Gate
5. XOR Gate
6. XNOR Gate
7. NOR Gate

} Universal gate

1. AND Gate :- In AND gate, whose output is high i.e when both the input are 1.
If a single input is 0 the output 0 will be obtained.

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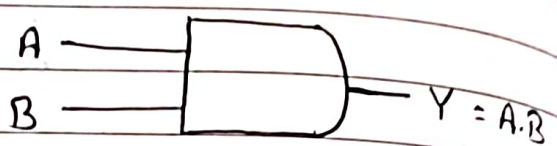
AND Gate Equation

$Y = A.B$

Circuit Diagram



Truth Table :

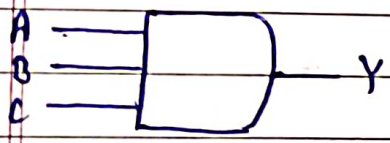
$Y = A.B$

A	B	$Y = A.B$
0	0	0
0	1	0
1	0	0
1	1	1

$Y = A.B.C$

A	B	C	$Y = A.B.C$
0	0	0	0
1	1	1	1
0	1	1	0
1	0	1	0
1	1	0	0
0	0	1	0
1	0	0	0
0	1	0	0

$Y = A.B.C$



2. OR Gate : This Gate has 2 or more input and one output. If a single Input 1 is high, the output 1 is obtained.

OR Gate Equation

$Y = A + B$

Circuit Diagram

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Truth Table

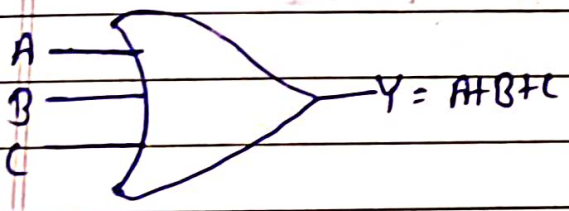
$Y = A+B$

A	B	Y = A+B
0	0	0
0	1	1
1	0	1
1	1	1

$Y = A+B+C$

A	B	C	Y = A+B+C
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$Y = A+B+C$



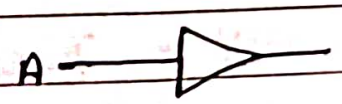
3. **NOT Gate** :- This is an inverting gate. It is also known as inverter

- It converts high input (1) to low (0) and vice versa.
- It has only one input and one output.
- The output of this gate is opposite of the input.

NOT Gate Equation

$Y = A'$

Circuit Diagram



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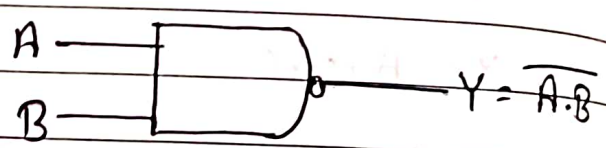
Truth Table

A	Y = A'
0	1
1	0

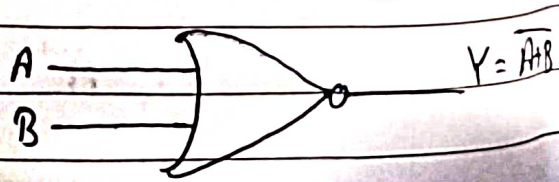
4 NAND Gate :- NOT + AND $(\overline{A.B})$

NAND Gate Equation	Circuit Diagram
$Y = \overline{A.B}$	

Truth Table

A	B	A.B	$\overline{A.B}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

5. NOR Gate :- NOT + OR $(\overline{A+B})$

NOR Gate Equation	Circuit Diagram
$Y = \overline{A+B}$	

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Truth Table

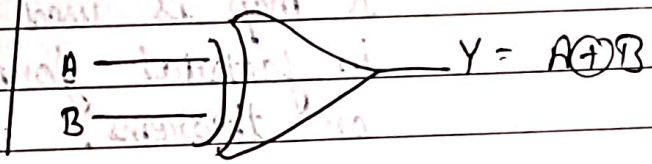
A	B	A+B	$\overline{A+B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

6. Exclusive OR (X-OR) ÷ [A ⊕ B]

X-OR Gate Equation

$Y = A \oplus B$

Circuit Diagram



Truth Table

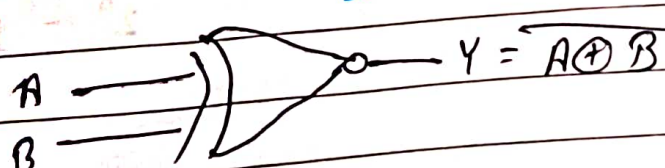
A	B	Y = A ⊕ B
0	0	0
0	1	1
1	0	1
1	1	0

7. Exclusive NOR (X-NOR) ÷ [$\overline{A \oplus B}$]

X-NOR Gate Equation

$Y = \overline{A \oplus B}$

Circuit Diagram



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Truth Table

A	B	$Y = \overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1

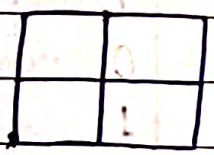
KAR NAUGH MAP (K-MAP)

K map is used to simplify the boolean expression in factorial format without using boolean laws or theorems.

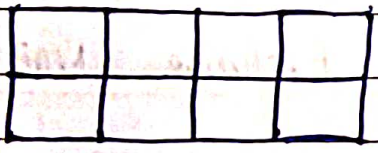
Note :- 1953 :- Developed by Karnaugh in the year 1953.

Formula :- 2^n = n is the no. of variables

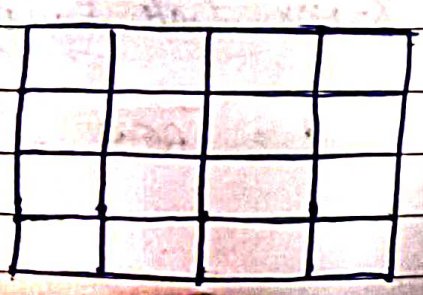
Ex :- $2^2 = 4$



$2^3 = 8$



$2^4 = 16$



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Keypoints

- $2^2 = 4$ cells

A \ B		
	0	1
	3	2

- $2^3 = 8$ cells

A \ BC				
	0	1	3	2
	4	5	7	6

- $2^4 = 16$ cells

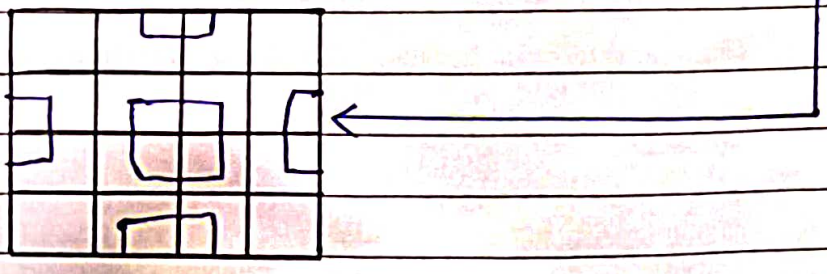
AB \ CD				
	0	1	3	2
	4	5	7	6
	12	13	15	14
	8	9	11	10

* No zeros allowed
* Group can be vertical or Horizontal but can not be diagonals.
* Over lapping allowed.
* Group should be as large as possible.
* Group must contain 2^n cells.

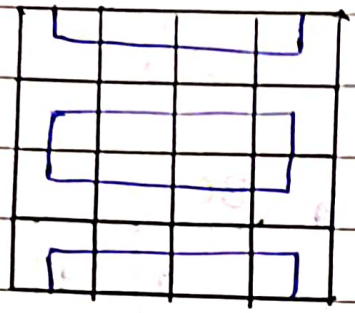
- PAIR GROUPS

* (2 cells)

- Quad (4 cells)



Oct groups (8 cells)



Minimization of Boolean Expressions :

Methods of Minimize Boolean Expressions

- By using Laws of Boolean Algebra
- By using K Maps

K Map (Karnaugh Map) : that provides a systematic method for by simplifying the boolean expressions.

For a boolean expression consisting of n-variables number of cells required in K Map = 2^n cells

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Two Variable K Map -

- Two variable K Map is drawn for a boolean expression consisting of two variables.
- The number of cells present in two variable K Map = 2^2 = 4 cells.
- So, for a boolean function consisting of two variables, we draw a 2 x 2 K Map.

Two variable K Map may be represented as -

A \ B	$\bar{B}$	B
$\bar{A}$	0	1
A	2	3

OR

A \ B	0	1
0	0	1
1	2	3

Three variable K Map -

- Three variable K Map is drawn for a boolean expression consisting of three variables.
- The number of cells present in three variable K Map = 2^3 = 8 cells
- So, for a boolean function consisting of three variables, we draw a 2 x 4 K Map.

Three variable K Map may be represented as -

A \ BC	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	1	3	2
A	4	5	7	6

OR

A \ BC	00	01	11	10
0	0	1	3	2
1	4	5	7	6

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Four variable K Map :

- four variable K Map is drawn for a boolean expression consisting of four variables.
- The number of cells present in four variable K Map = $2^4 = 16$ cells.
- So, for a boolean function consisting of four variables, we draw a 4 × 4 KMap.

- Four variable K Map may be represented as :

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	1	3	2
$\bar{A}B$	4	5	7	6
AB	12	13	15	14
$A\bar{B}$	8	9	11	10

OR

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

K Map Simplification Rules :

To minimize the given boolean function,

- We draw a K Map according to the number of variables it contains.
- We fill the K Map with 0's and 1's according to it's function.
- Then, we minimize the function in accordance with the following rules.
- A will use when only one ≠ map is using in pairing out of both.

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Rule ÷ 01

- We can either group 0's with 0's or 1's with 1's but we can not group 0's and 1's together.
- X representing don't care can be grouped with 0's as well as 1's.

Note ÷

- There is no need of separately grouping X's ie. they can be ignored if all 0's and 1's are already grouped.

Rule ÷ 02

- Group may overlap each other.

Rule ÷ 03

- We can only create a group whose number of cells can be represented in the power of 2.
- In other words, a group can only contain 2^n i.e. 1, 2, 4, 8, 16 and so on number of cell

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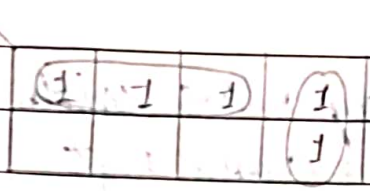
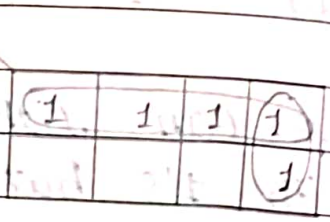
Ex:-



Incorrect (X)

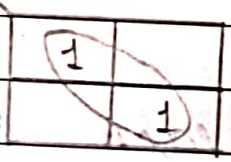


Correct (✓)

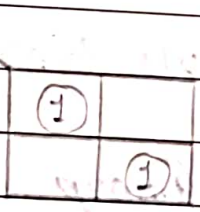
Rule :- 04

- Groups can only either horizontal or vertical.
- We can not create groups of diagonal or any other shape.

Ex :-



Incorrect (x)



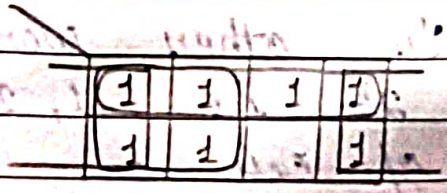
Correct (✓)

Rule :- 05

- Each group should be as large as possible.

Ex :-

Incorrect (x)



Correct (✓)

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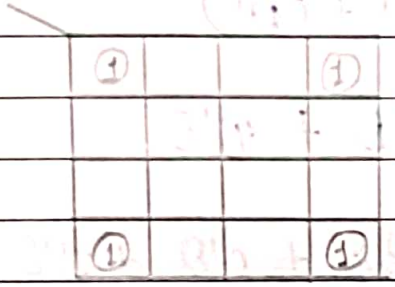
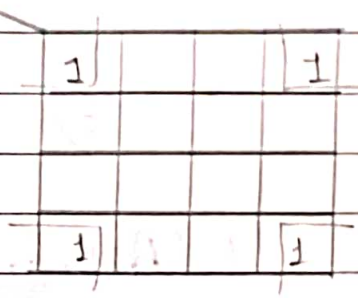
Rule ÷ 06

- Opposite grouping and corner grouping are allowed.
- The example of opposite grouping is shown illustrated in Rule - 05
- The example of corner grouping is show below.

Ex ÷



Incorrect (X)



Correct (✓)

Rule ÷ 07

- There should be as few groups as possible.

Problem Based ON Karnaugh Map ÷

Problem 01 ÷ Minimize the following boolean function -

$F(A, B, C, D) = \Sigma m (0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$

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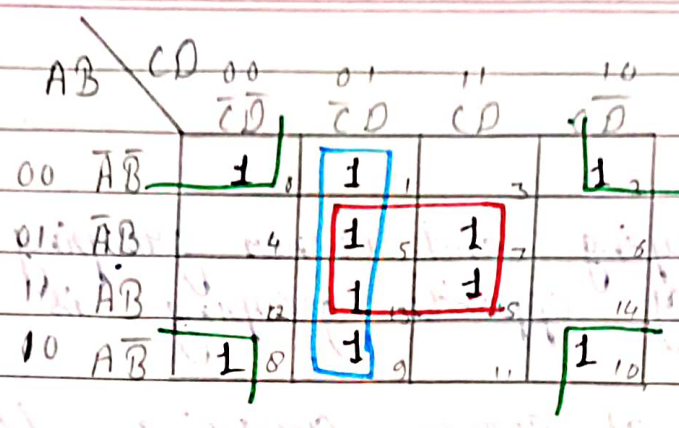
Solution:



$F(A,B,C,D) \Rightarrow (A'B + AB)(C'D + (D) + (A'B' + A'B + AB + AB')(C'D + (A'B' + AB')(C'D' + (D')$

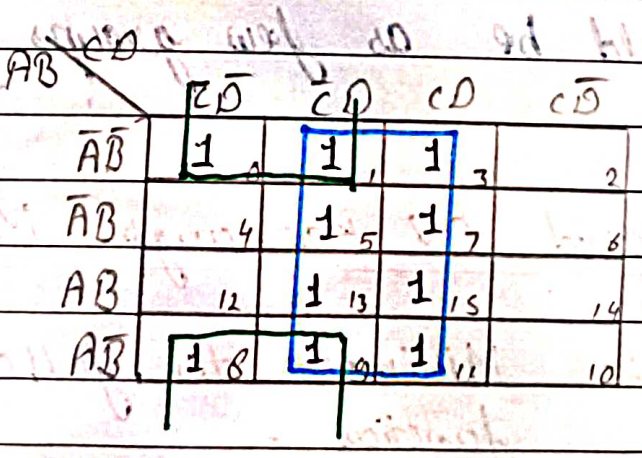
$\Rightarrow BD + C'D + B'D'$

$F(A,B,C,D) = BD + C'D + B'D$ ∴

Problem .02 :- Minimize the following boolean function.

$F(A,B,C,D) = \Sigma m(0,1,3,5,7,8,9,11,13,15)$

Solution:-



$F(A,B,C,D) \Rightarrow (A'B' + A'B + AB + AB')(C'D + (D) + (A'B')(C'D' + (C'D) + (A'B')(C'D' + C'D)$

$F(A,B,C,D) \Rightarrow D + B'C'$ ∴

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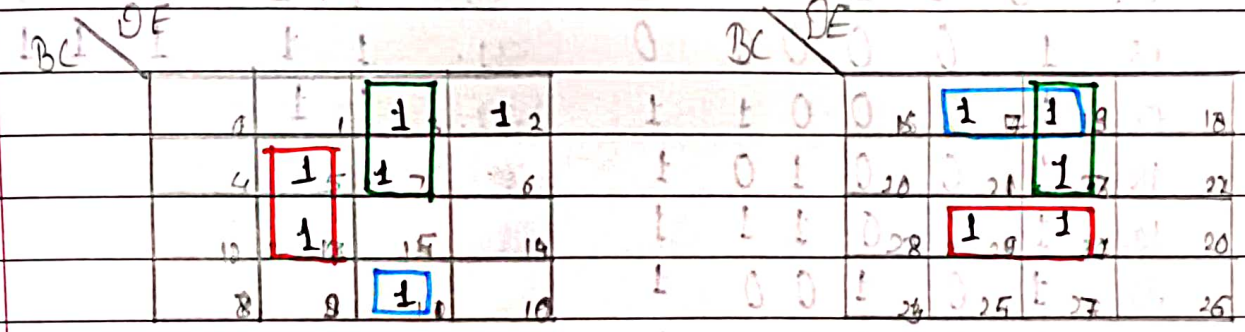
Problem - 03 :- Design a circuit which has five bit binary in code (A, B, C, D, E) with A Being MSB, It must produce an output logic high for any prime number detected in the input data.

Solution. $f(A,B,C,D,E) \Rightarrow \Sigma m\,(2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31)$



$f(A,B,C,D,E) \Rightarrow \bar{B}DE + \bar{A}C\bar{D}E + ABCE + A\bar{B}\bar{C}E + \bar{A}\bar{B}\bar{C}D + AB\bar{C}DE$

Problem - 04 :- Design a circuit of five input variables that generate output 1 if and only if the number of 1's in the input is prime.

Solution :-

	2^4 A	2^3 B	2^2 C	2^1 D	2^0 E	Y	
1.	0	0	0	0	0	0	Not Prime
2.	0	0	0	0	1	0	"
3.	0	0	0	1	1	1	"
4.	0	0	1	0	0	0	
5.	0	0	1	0	1	1	⇒ Prime
6.	0	0	1	1	0	1	

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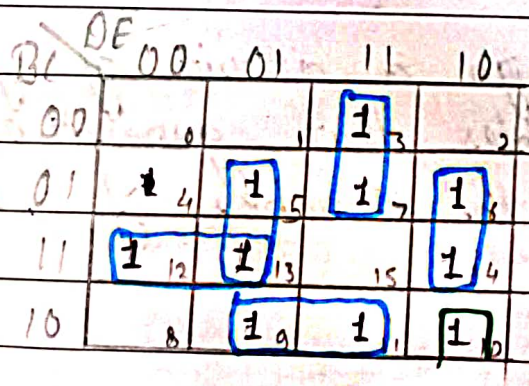
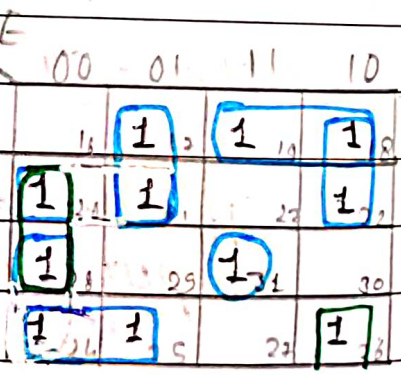
	2⁴	2³	2²	2¹	2⁰	Y
7.	0	0	1	1	1	1
8.	0	1	0	0	0	0
9.	0	1	0	0	1	1
10.	0	1	0	1	0	1
11.	0	1	0	1	1	1
12.	0	1	1	0	0	1
13.	0	1	1	0	1	1
14	0	1	1	1	0	1
15.	0	1	1	1	1	0
16.	1	0	0	0	0	0
17.	1	0	0	0	1	1
18.	1	0	0	1	0	1
19.	1	0	0	1	1	1
20.	1	0	1	0	0	1
21.	1	0	1	0	1	1
22.	1	0	1	1	0	1
23.	1	0	1	1	1	0
24.	1	1	0	0	0	1
25.	1	1	0	0	1	1
26.	1	1	0	1	0	1
27.	1	1	0	1	1	0
28.	1	1	1	0	0	1
29.	1	1	1	0	1	0
30.	1	1	1	1	0	0
31.	1	1	1	1	1	1
32.	1	1				

SOP form for $f(A,B,C,D,E) \Rightarrow \Sigma m(3,5,6,7,9,10,11,12,13,14,17,18,19,20,21,22,24,25,26,28,31)$





Expression: $f(A,B,C,D,E) \Rightarrow$

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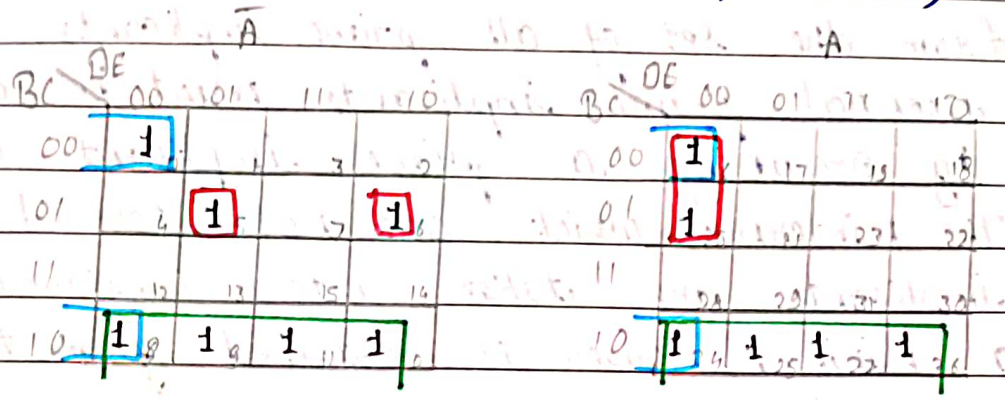
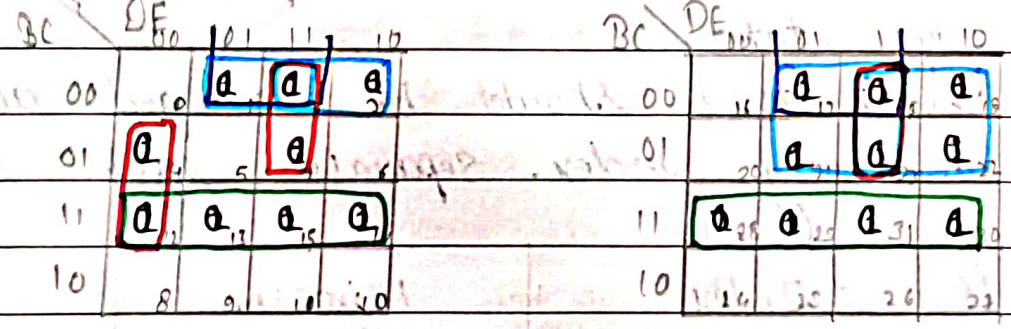
Problem - 04 : Simplify

SOP form :- $f(A, B, C, D, E) \Rightarrow \Sigma m\,(0, 5, 6, 8, 9, 10, 11, 16, 20, 24, 25, 26, 27)$



Expression $f(A, B, C, D, E) \Rightarrow B\bar{C} + C\bar{D}\bar{E} + A\bar{B}\bar{D}\bar{E} + A\bar{B}\bar{C}\bar{D}E + \bar{A}\bar{B}CD\bar{E}$

POS form :- (Max term)



Expression $f(A, B, C, D, E) \Rightarrow (\bar{B}+\bar{C})(\bar{A}+B+\bar{D})(\bar{A}+B+\bar{E})(A+\bar{B}+\bar{D}+\bar{E}) + (B+C+\bar{E})(A+\bar{C}+D+E)$

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Quine Mcclustey Method (Tabular Method)

Step 1. A set of all prime implicants of the function must be obtained.

Step 2. From the set of all prime implicants a set of essential prime implicants must be determined by preparing a prime implicant chart.

Step 3. The minterms which are not covered by essential implicants are taken into consideration and a minimum cover is obtained from the remaining prime implicants.

Selecting prime implicants.

Step 1. Each minterm should be expressed by its binary representation.

Step 2. The minterms should be arranged according to increasing index. Separate each set of minterm

i.e.:

Index (No. of 1's)	Minterm	Variables A B C D
0	0	0 0 0 0
1	1	0 0 0 1
1	2	0 0 1 0
2	3	0 0 1 1

- possesing the same index by lines.

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Step 3. Each term of index 'n' should be compared with each term of index 'n+1', for each pair of term that can combine the newly formed ~~i.e.~~ terms should be stated. If two minterm ~~differ~~ differs in only 1 variable that variable should be removed and Thus a new term with one less literal is form after all the pairs of terms with index 'n' and 'n+1' have been consider a line should be drawn under the last term.

Step 4. When the above process has been completed for all the groups of terms, 1 stays of elemination will be completed.

Step 5. The next stage of elemination should be repeated for the new term, according to this stage the two term only combine when they have dashes in only same position.

Step 6. The cycles have to be continue, until no new list can be found.

Step 7. All terms which remain unticked during the process are prime implicants.

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Prime Implicant Chart

Step 1. The prime implicant should be represented in rows and minterm should be in column.

Step 2. Process should be placed in each row to show the composition of minterms that makes the prime.

Step 3. A completed prime implicant table should be inspected for columns containing only single cross. prime implicant that cover minterms with a single cross are called essential prime implicant.

Problem 1 :- $f(A,B,C,D) = \Sigma m(1,2,3,7,8,9,10,11,14,15)$

Solution :- • Binary Representation.

Minterms		Variables A B C D
1	→	0 0 0 1
2	→	0 0 1 0
3	→	0 0 1 1
7	→	0 1 1 1
8	→	1 0 0 0
9	→	1 0 0 1
10	→	1 0 1 0
11	→	1 0 1 1
14	→	1 1 1 0
15	→	1 1 1 1

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• Arrangement acc. to Index.

Index (No of 1's)	Minterm	Variable A B C D
1	1	0 0 0 1 ✓
	2	0 0 1 0 ✓
	8	1 0 0 0 ✓
2	3	0 0 1 1 ✓
	9	1 0 0 1 ✓
	10	1 0 1 0 ✓
3	7	0 1 1 1 ✓
	11	1 0 1 1 ✓
	14	1 1 1 0 ✓
4	15	1 1 1 1

• Compare 'n' with 'n+1'

n.

Index (No of 1's)	Minterm	Variable A B C D
1	1, 3	0 0 _ 1 ✓
	1, 9	_ 0 0 1 ✓
	2, 3	0 0 1 _ ✓
	2, 10	_ 0 1 0 ✓
	8, 9	1 0 0 _ ✓
	8, 10	1 0 _ 0 ✓

n+1.

2	3, 7	0 _ 1 1 ✓
	3, 11	_ 0 1 1 ✓
	9, 11	1 0 _ 1 ✓
	10, 11	1 0 1 _ ✓
	10, 14	1 _ 1 0 ✓

n+2	3.	7, 15	- 1 1 1 ✓
		11, 15	1 - 1 1 ✓
		14, 15	1 1 1 - ✓

Compare Index 'n' with 'n+1'

Index	Minterm	Variable (A B C D)	
n 1	1, 3, 9, 11	- 0 - 1	$\bar{B}D$
	1, 9, 3, 11	- 0 - 1	$\bar{B}D$
	2, 3, 10, 11	- 0 1 -	$\bar{B}C$
	2, 10, 3, 11	- 0 1 -	$\bar{B}C$
	8, 9, 10, 11	1 0 - -	$A\bar{B}$
	8, 10, 9, 11	1 0 - -	$A\bar{B}$
n+1 2	3, 7, 11, 15	- - 1 1	CD
	3, 11, 7, 15	- - 1 1	CD
	10, 11, 14, 15	1 - 1 -	AC
	10, 14, 11, 15	1 - 1 -	AC

Prime Implicant chart :-

8 Prime Implicants

Prime Implicants	1	2	3	7	8	9	10	11	14	15
$\bar{B}D$	(X)		X			X	X	X		
$\bar{B}C$		(X)	X				X	X		
$A\bar{B}$					(X)	X	X	X		
CD			X	(X)				X		X
AC							X	X	(X)	X

$f(A,B,C,D) = \bar{B}D + \bar{B}C + CD + A\bar{B} + AC$ Ans

Problem 2 :- $f(w,x,y,z) = f(w,x,y,z) \geq m(1,3,4,5,9,10,11) + \geq(6,8)$

Solution :- • Binary representation of all minterms and don't care:

Minterms	Variables
1	0 0 0 1
3	0 0 1 1
4	0 1 0 0
5	0 1 0 1
6	0 1 1 0
8	1 0 0 0
9	1 0 0 1
10	1 0 1 0
11	1 0 1 1

Arrangement according to index.

Index- No. of 1's	Minterm	Variable (W X Y Z)	
1	1	0 0 0 1	✓
	4	0 1 0 0	✓
	8	1 0 0 0	
2	3	0 0 1 1	✓
	5	0 1 0 1	✓
	6	0 1 1 0	✓
	9	1 0 0 1	✓
	10	1 0 1 0	✓
3	11	1 0 1 1	✓

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• Compare 'n' with 'n+1'

Index No. of 1's	Minterm	Variable
1	1,3	0 0 - 1
	1,5	0 - 0 1
	1,9	- 0 0 1
	4,5	0 1 0 -
	4,6	0 1 - 0
	8,9	1 0 0 -
	8,10	1 0 - 0
2	3,11	- 0 1 1
	10,11	1 0 1 -
	9,11	1 0 - 1

• Compare index 'n' with 'n+1' again.

Index No. of 1's	Minterm	Variable	
1	1,3,9,11	- 0 - 1	$\bar{B}D = \bar{X}Z$
	1,9,3,11	- 0 - 1	$\bar{B}D = \bar{X}Z$
	8,9,10,11	1 0 - -	$A\bar{B} = W\bar{X}$
	8,10,9,11	1 0 - -	$A\bar{B} = W\bar{X}$

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• Prime Implicant Chart :-

Prime Implicants	1	3	4	5	9	10	11
1,5	X			X			
4,5			X	X			
4,6			X				
1,3,9,11	X	(X)			X		X
8,10,9,11					X	(X)	X

$\Rightarrow \bar{X}Z + W\bar{X}$

• Prime Implicant chart for left minterm [4,5]

Prime Implicants	1	3	4	5	9	10	11
1,5				X			
4,5			(X	X)			
4,6			X				

$f(W,X,Y,Z) \Rightarrow \bar{X}Z + W\bar{X} + \bar{W}X\bar{Y}$

Problem 03:- using the quine Mccluskey method and K-map Simplify the following function.

$F(W,X,Y,Z) = \Sigma m(0,2,5,7,9,11,13,15,16,18,21,23,25,27,29,31)$

Solution :-

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- Binary Representation :

Minterms	Variables				
	U	W	X	Y	Z
0	0	0	0	0	0
2	0	0	0	1	0
5	0	0	1	0	1
7	0	0	1	1	1
9	0	1	0	0	1
11	0	1	0	1	1
13	0	1	1	0	1
15	0	1	1	1	1

Minterms	Variable				
	U	W	X	Y	Z
16	1	0	0	0	0
18	1	0	0	1	0
21	1	0	1	0	1
23	1	0	1	1	1
25	1	1	0	0	1
27	1	1	0	1	1
29	1	1	1	0	1
31	1	1	1	1	1

- Arrangement acc. to index.

Index No. of 1's	Minterm	Variable				
0	0	0	0	0	0	0
1	2	0	0	0	1	0 ✓
	16	1	0	0	0	0 ✓
2	5	0	0	1	0	1
	9	0	1	0	0	1
	18	1	0	0	1	0
3	7	0	0	1	1	1
	11	0	1	0	1	1
	13	0	1	1	0	1
	21	1	0	1	0	1
	25	1	1	0	0	1

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4	15	0 1 1 1 1
	23	1 0 1 1 1
	27	1 1 0 1 1
	29	1 1 1 0 1
5	31	1 1 1 1 1

Compare 'n' with 'n+1' (1st)

Index No. of 1's	Minterms	Variables
0	0, 2	0 0 0 - 0
	0, 16	- 0 0 0 0
1	2, 18	- 0 0 1 0
	16, 18	1 0 0 - 0
2	5, 7	0 0 1 - 1
	5, 13	0 - 1 0 1
	5, 21	- 0 1 0 1
	9, 11	0 1 0 - 1
	9, 25	- 1 0 - 1
	9, 13	0 1 - 0 1
3	7, 15	0 - 1 1 1
	7, 23	- 0 1 1 1
	13, 15	0 1 1 - 1
	11, 15	0 1 - 1 1
	11, 27	- 1 0 1 1
	13, 29	- 1 1 0 1
	21, 23	1 0 1 - 1
	25, 27	1 1 0 - 1
	25, 29	1 1 - 0 1
	21, 29	1 - 1 0 1

3	4	15, 31	- 1 1 1 1
		23, 31	1 - 1 1 1
		27, 31	1 1 - 1 1
		29, 31	1 1 1 - 1

Compare 'n' with 'n+1' again (2nd)

Index No. of 1's	Minterms	Variables
0	0, 2, 16, 18	- 0 0 - 0
	0, 16, 2, 18	- 0 0 - 0
2		

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Q.2 Reduce the following using Kmap Technique

$Y(A,B,C,D) = \pi m\ (0,1,2,3,8,9,10,11,12,13)$

Solution :-

POS Form

AB \ CD	00	01	11	10
A+B (00)	0 (0)	0 (1)	0 (3)	0 (2)
A+B̄ (01)	4	5	7	6
Ā+B̄ (11)	0 (12)	0 (13)	15	14
Ā+B (10)	0 (8)	0 (9)	0 (11)	0 (10)

Expression :- $Y(A,B,C,D) = B.(\bar{A}+C)$ Ans

SOP Form

AB \ CD	00	01	11	10
ĀB̄	0	1	3	2
ĀB	1 (4)	1 (5)	1 (7)	1 (6)
AB	12	13	1 (15)	1 (14)
AB̄	8	9	11	10

Expression :- $Y(A,B,C,D) = \bar{A}B + BC$ Ans

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Question 3 :- Write Draw the NAND Implimentation of the fallowing.

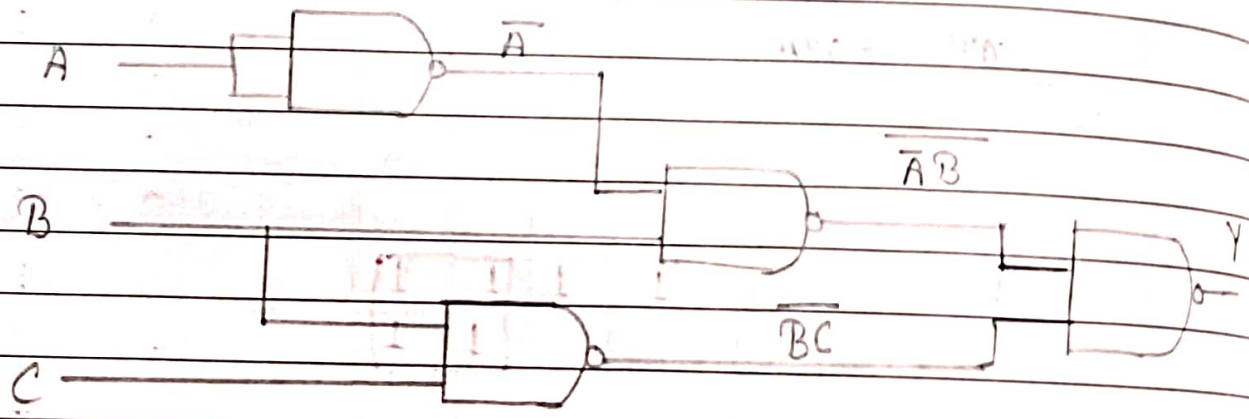
$$Y = \bar{A}B + BC$$

Solution.

$$Y = \bar{A}B + BC$$

$$Y = \overline{\bar{A}B + BC}$$

$$Y = \overline{(\bar{A}B)} \cdot \overline{(BC)}$$



$$Y = \overline{(\bar{A}B)} \cdot \overline{(BC)}$$

Question 4 :- Draw the NOR Implimentation of the fallowing

$$Y = B \cdot (\bar{A} + C)$$

Solution,

$$Y = \overline{B \cdot (\bar{A} + C)}$$

$$Y = \overline{\bar{B} + (\bar{A} + C)}$$

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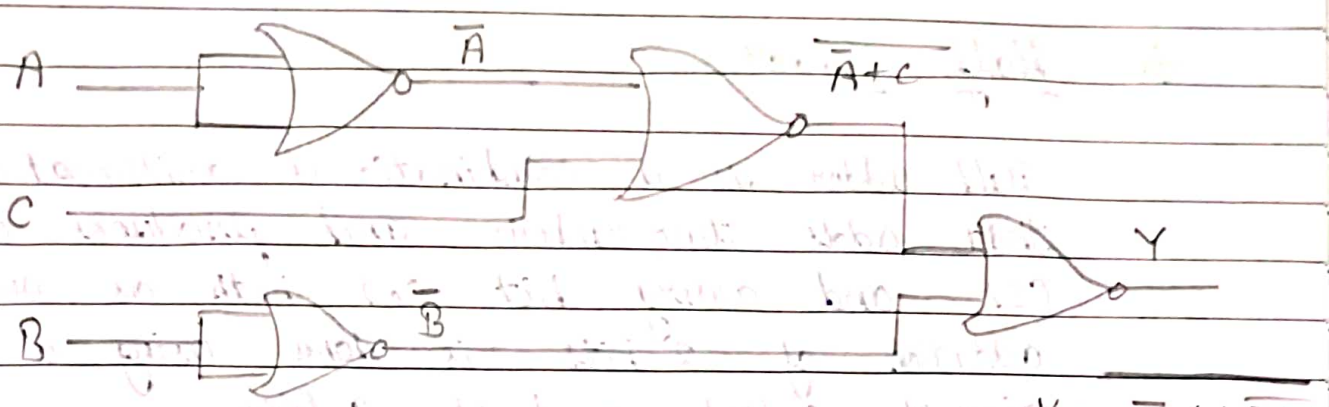
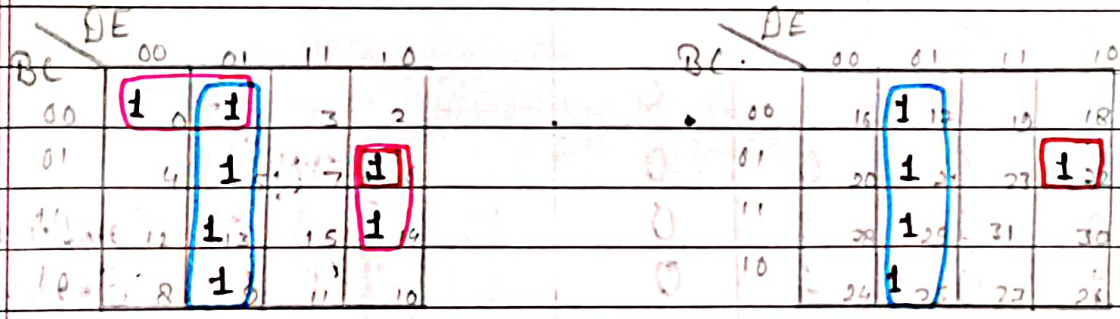


$Y = \bar{B} + (\bar{A}+C)$

Question 5 :- Find out the expression for

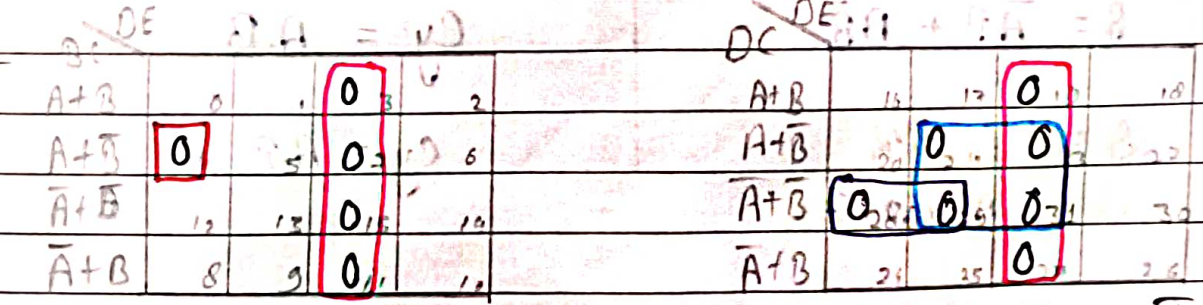
$Y(A,B,C,D,E) \Rightarrow \Sigma m\,(0,1,5,6,9,13,14,17,21,22,25,29)$



Expression :- $Y(A,B,C,D,E) = \bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{B}CD\bar{E} + \bar{A}CD\bar{E}$

Question 6 :- Function is $Y(A,B,C,D,E) = \pi M\,(3,4,7,11,15,19,21,23,27,28,29,31)$

POS form



Expression :- $Y(A,B,C,D,E) = (\bar{D}+\bar{E}).(\bar{A}+\bar{C}+\bar{E}).(\bar{A}+\bar{B}+\bar{C}+D).(A+B+\bar{C}+D+E)$

Combinational Circuit

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★ Half Adder

Half adder is a combinational arithmatic circuit that adds two numbers and produces a sum bit (S) and carry bit (c) both as output. The addition of 2 bits is done using a combination circuit called a half adder.

- Combinational arithmatic circuit

Truth Table

Input		Output	
A	B	S	Cy
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

- Sum = S
- Carry = Cy

2 Types
- Combinational
- Sequential

Expression with the help of K-Map

A \ B	0	1
0	0	1
1	1	0

$S = \bar{A}B + A\bar{B}$

$S = A \oplus B$

A XOR B

A \ B	0	1
0	0	0
1	0	1

$Cy = A.B$

$Cy = A.B$

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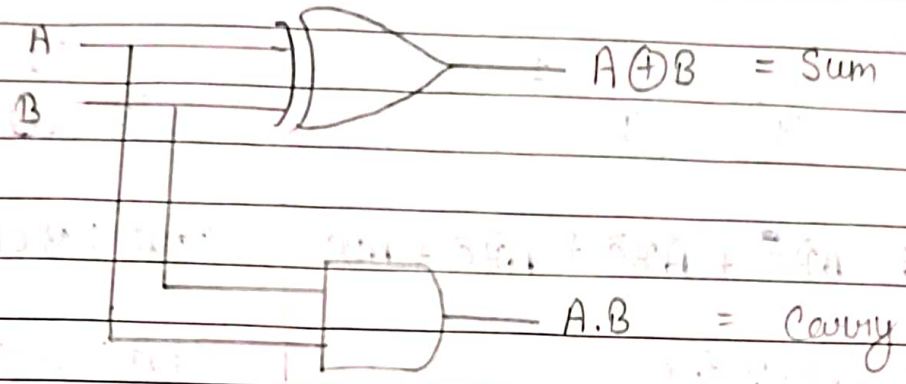
Impliment for XOR and NAN: AND Expressions



★ Full Adder

Full Adder is the adder that adds three inputs and produces two outputs.

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

⇒ The first two inputs are A and B and the third input is an input carry as C-IN.

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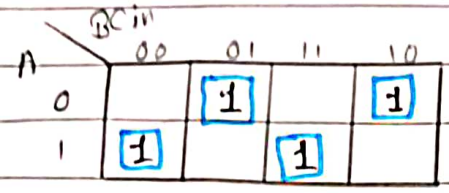
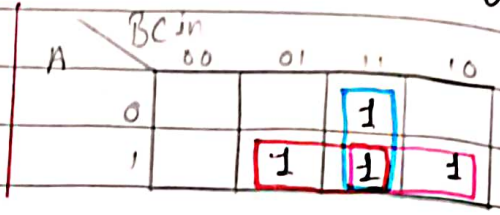
Expression of the table with the help of K-Map





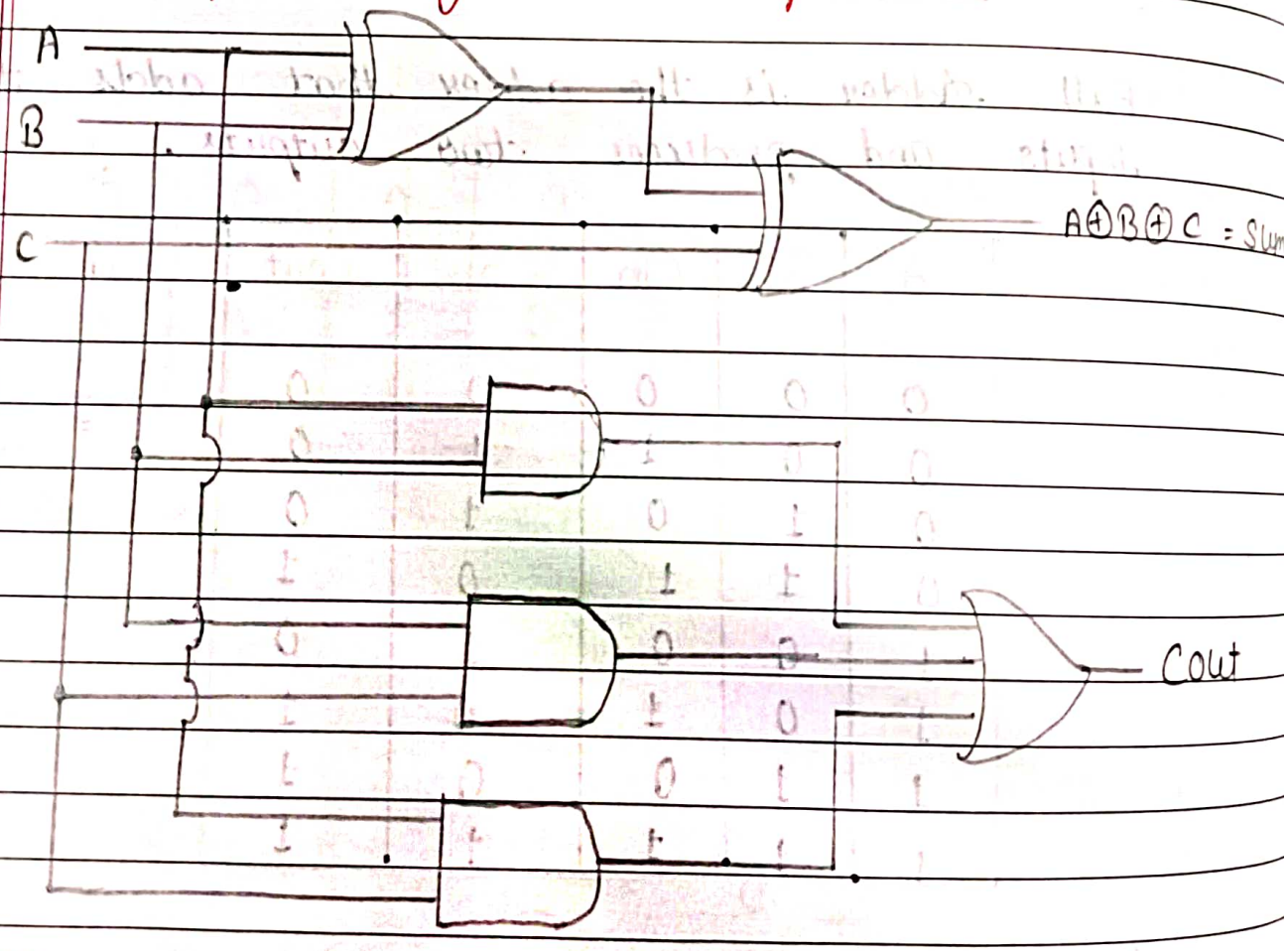
Sum ÷ $\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$

$S = A \oplus B \oplus C$

Cout ÷ BCin + ACin + AB

Cout ÷ BCin + Acin + AB

Impliment for the expressions.



LSB = Least Significant Bit

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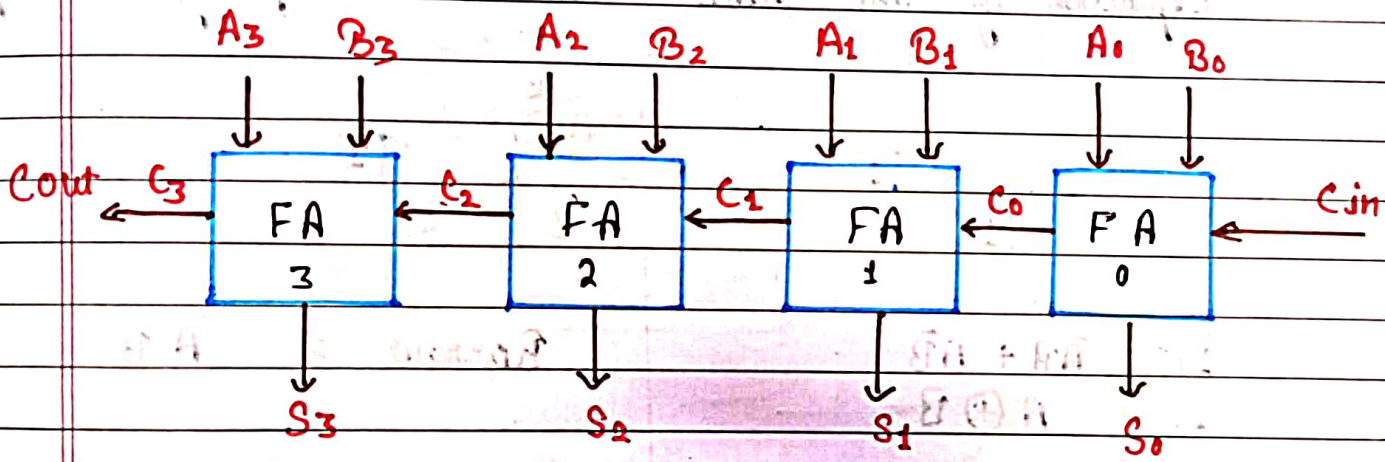
Binary Adder

Binary Adders are arithmetic circuits in the form of half-adders and full-addersb used to add . . together two binary digits.

The Binary Adder is formed with the help of the Full-Adder circuit. The Full-Adders are connected in series, and the output carry of the first Adder will be treated as the input carry of the next Full-Adder.

4-Bit Binary Adder



This is binary adder for :-

$$\begin{array}{r} A_3\ A_2\ A_1\ A_0 \\ +\ B_3\ B_2\ B_1\ B_0 \\ \hline \end{array}$$

=> The outcome produced by adding both input variables is stored into either third register or to one of the source registers.

∴ FA :- full Adder

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Half Subtractor

A half - subtractor is a combinational logic circuit that have two inputs and two outputs.

A	B	D	Borrow
0	0	0	0
0	1	1	1
1	1	0	0
1	0	1	0

Expression of the table with the help of K-Map

A \ B	0	1
0	0	1
1	1	0

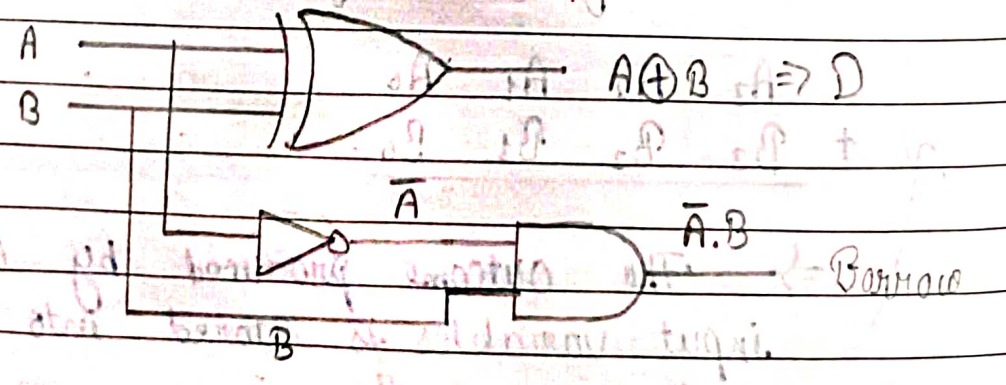
A \ B	0	1
0	0	1
1	0	0

$D \Rightarrow \bar{A}B + A\bar{B}$

$D \Rightarrow A \oplus B$

Borrow $\Rightarrow \bar{A}.B$

Impliment for the expressions.



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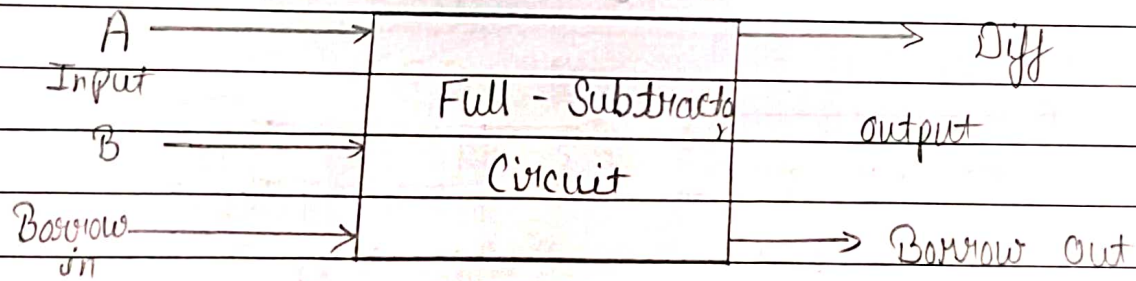
Full Subtractor

The Half Subtractor is used to subtract only two numbers. To overcome this problem, a full subtractor was designed. The full subtractor is used to subtract three 1-bit numbers A, B and C, which are minuend, subtrahend, and borrow, respectively. The full subtractor has three input states and two output states ~~or~~ i.e, diff and borrow.

Block diagram

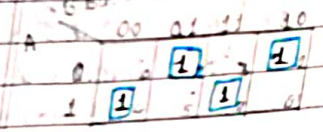


Truth Table

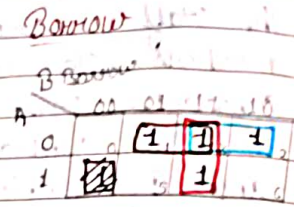
Inputs			Output	
A	B	Borrow in	Diff	Borrow
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Expression of the table with the help of K-map

Diff :



Borrow

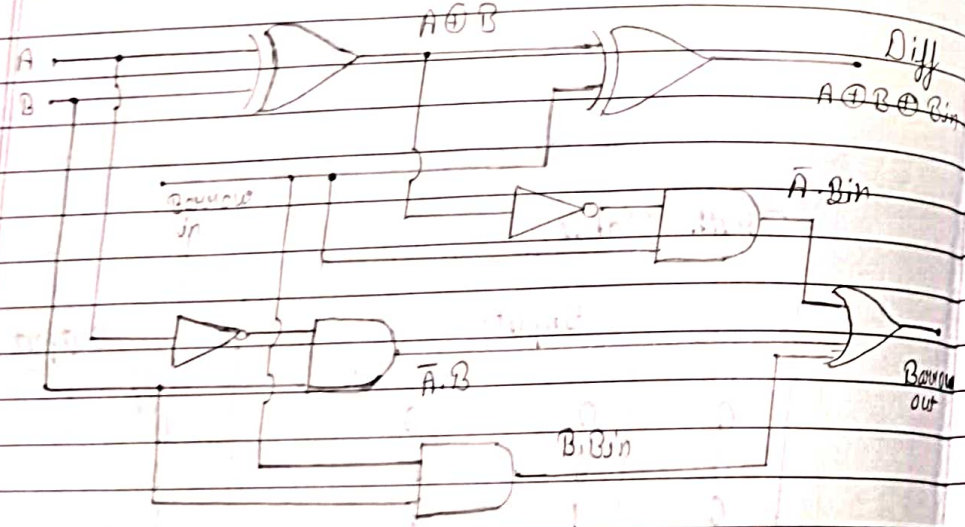


Diff = $A\bar{B}\bar{B}_{in} + \bar{A}\bar{B}B_{in} + AB B_{in} + \bar{A}B\bar{B}_{in}$.

Diff = $A \oplus B \oplus B_{in}$

Borrow = $\bar{A}.B_{in} + \bar{A}B + B.B_{in}$

Construction of full subtractor circuit :



OR =

Binary Subtractor

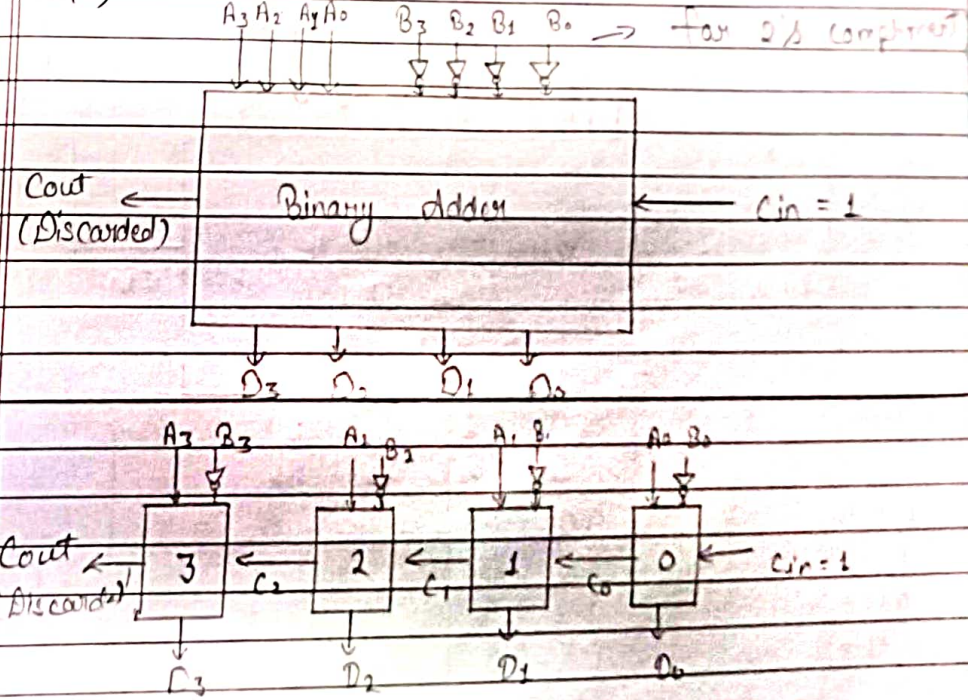
As their name implies, a Binary subtractor is a decision making circuit that subtracts two binary numbers from each other, for example X-Y to find the resulting difference between the two numbers.

Let Binary Subtractor for,

$A_3\ A_2\ A_1\ A_0$

$-\ B_3\ B_2\ B_1\ B_0$

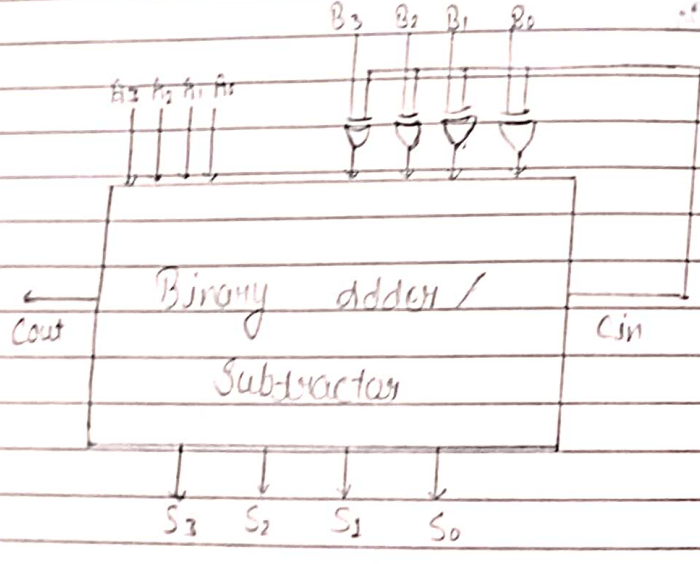
Ex,



☆ Binary adder / subtractor

In digital circuits, A Binary Adder- Subtractor is capable of both the addition and subtraction of binary numbers in one circuit itself. The operation is performed depending on the binary value the control signal holds. It is one of the components of the ALU (Arithmetic Logic Unit)

To work with Binary Adder- Subtractor it is required that we have Knowledge of the XOR gate, Full- Adder, Binary Addition and Subtraction.



↗ ADD

0 0 → 0
1 0 → 1

0 1 → 1

☆ BCD [Binary Coded Decimal]

Binary coded Decimal, or BCD, is another process for converting decimal numbers into their binary equivalents.

- It is a form of binary encoding where each digits in a decimal number is represented in the form of bits.
- This encoding can be done in 4-bit or 8-bit (usually 4-bit is preferred).
- It is a fast and efficient system that converts the decimal numbers into the binary numbers as compared to the existing binary system.
- These are generally used in digital displays where is the manipulation of data is quite a task.
- Thus BCD plays an important role here bcz the manipulation is done treating each digit as a separate single sub-circuit.

Truth Table for Binary Coded Decimal

Decimal Number	BCD
0	0 0 0 0
1	0 0 0 1
2	0 0 1 0
3	0 0 1 1
4	0 1 0 0

D.N	BCD
6	0 1 1 0
7	0 1 1 1
8	1 0 0 0
9	1 0 0 1

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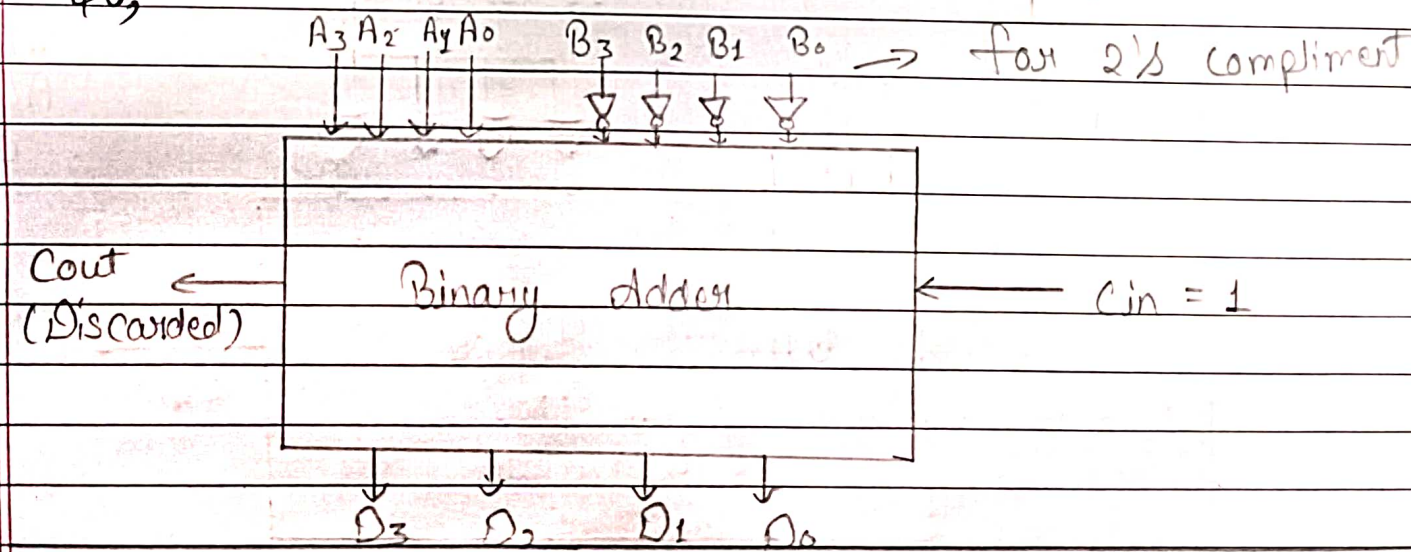
Binary Subtractor

As their name implies, a Binary subtractor is a decision making circuit that subtracts two binary numbers from each other, for example X - Y to find the resulting difference between the two numbers.

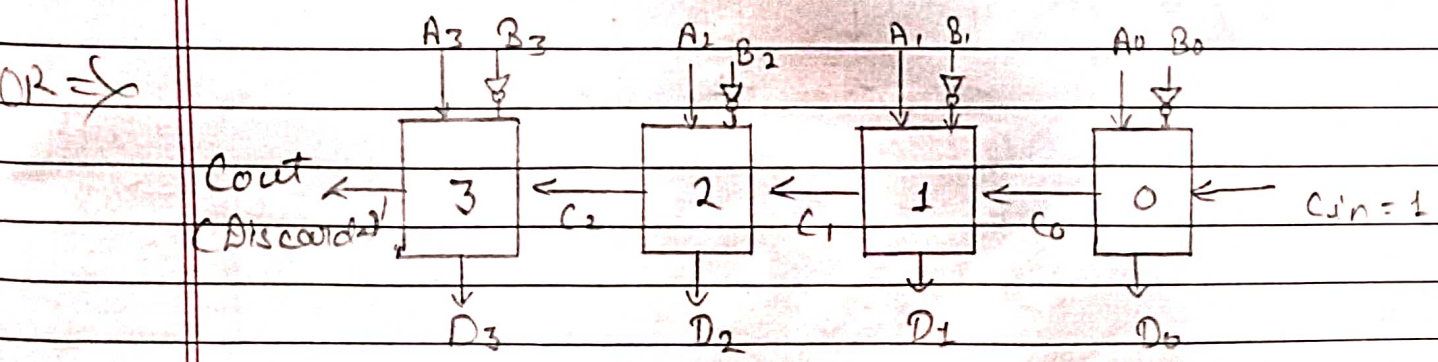
Let Binary Subtractor for,

$$\begin{array}{cccc} A_3 & A_2 & A_1 & A_0 \\ - \; B_3 & B_2 & B_1 & B_0 \\ \hline \end{array}$$

So,



OR =>



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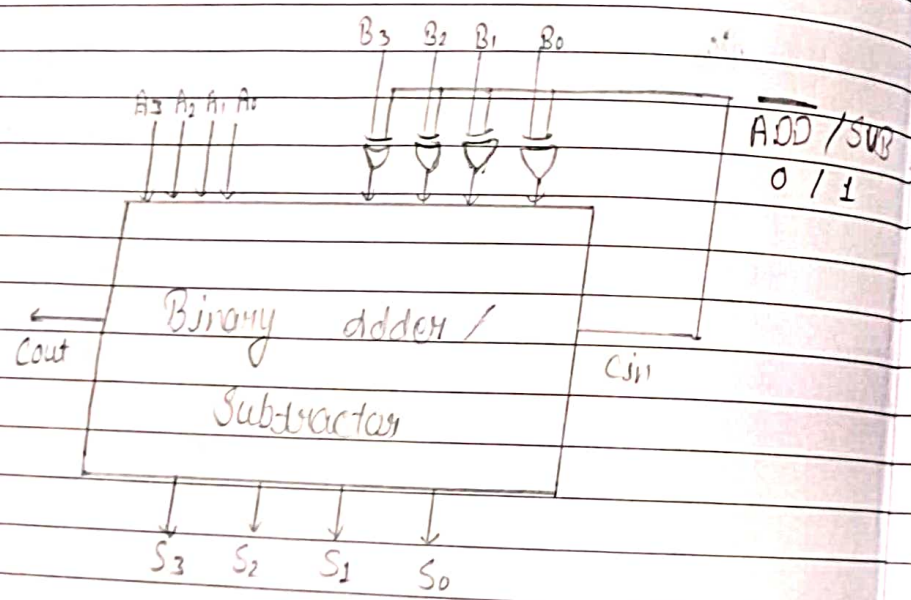

↗ ADD

0 0 → 0
1 0 → 1

0 1 → 1
1 1 → 0

Subtract

★ BCD [Binary Coded Decimal]

Binary coded Decimal, or BCD, is another process for converting decimal numbers into their binary equivalents.

- It is a form of binary encoding where each digits in a decimal number is represented in the form of bits.
- This encoding can be done in 4-bit or 8-bit (Cusually 4-bit is preferred).
- It is a fast and efficient system that converts the decimal numbers into the binary numbers as compared to the existing binary system.
- These are generally used in digital displays where is the manipulation of data is quite a task.
- Thus BCD plays an important role here bcz the manipulation is done treating each digit as a separate single sub-circuit.

Truth Table for Binary Coded Decimal.

Decimal Number	BCD
0	0 0 0 0
1	0 0 0 1
2	0 0 1 0
3	0 0 1 1
4	0 1 0 0
5	0 1 0 1

D.N	BCD
6	0 1 1 0
7	0 1 1 1
8	1 0 0 0
9	1 0 0 1

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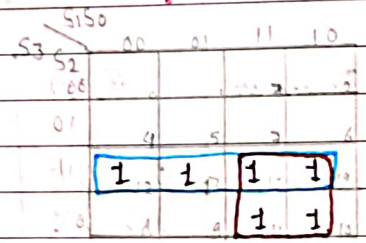
Decimal Adder (BCD Adder)

The BCD- Adder is used in the computers and the calculators that perform arithmetic operation directly in the decimal number system.

valid				Invalid			
~~A_3~~ S_3	~~A_2~~ S_2	~~A_1~~ S_1	~~A_0~~ S_0	~~A_3~~ S_3	A_2 S_2	~~A_1~~ S_1	~~A_0~~ S_0
0	0	0	0	1	0	1	0
0	0	0	1	1	0	1	1
0	0	1	0	1	1	0	0
0	0	1	1	1	1	0	1
0	1	0	0	1	1	1	0
0	1	0	1	1	1	1	1
0	1	1	0				
0	1	1	1				
1	0	0	0				
1	0	0	1				

- K-Map for Invalid BCD



$$F = S_3S_2 + S_3S_1$$

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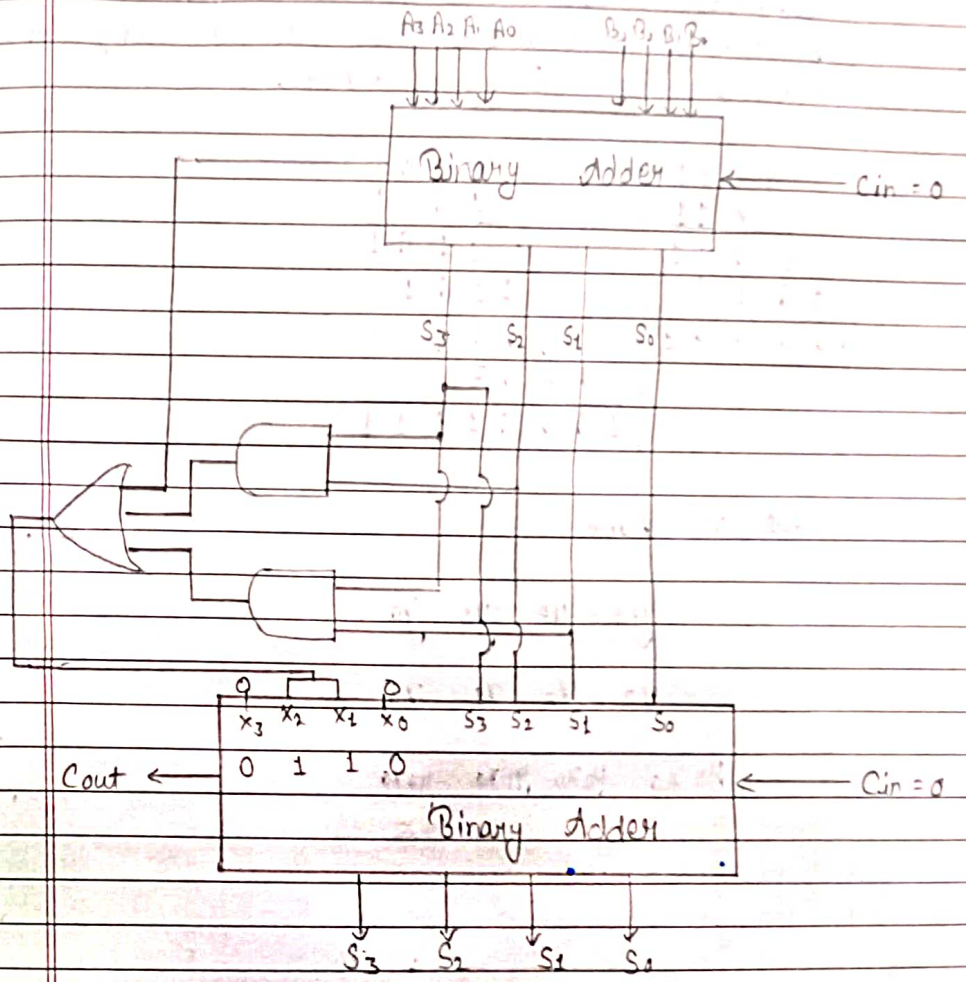
Circuit diagram for Decima adder



* Note :- The decimal number requires 4 bits to represent in the BCD code. and the circuit must have an input carry and an output carry.

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★ Binary Multiplier

A binary multiplier is an electronic circuit used in digital electronics, such as a computer, to multiply two binary numbers.

1011	1101
× 11	× 1011
1011	1101
1011×	1101××
100001	0000×××
	1101××××
	10001111

Let, we have

$$y_3\ y_2\ y_1\ y_0$$
$$\times\ x_3\ x_2\ x_1\ x_0$$

y_3x_0 y_2x_0 y_1x_0 y_0x_0

y_3x_1 y_2x_1 y_1x_1 y_0x_1 ×

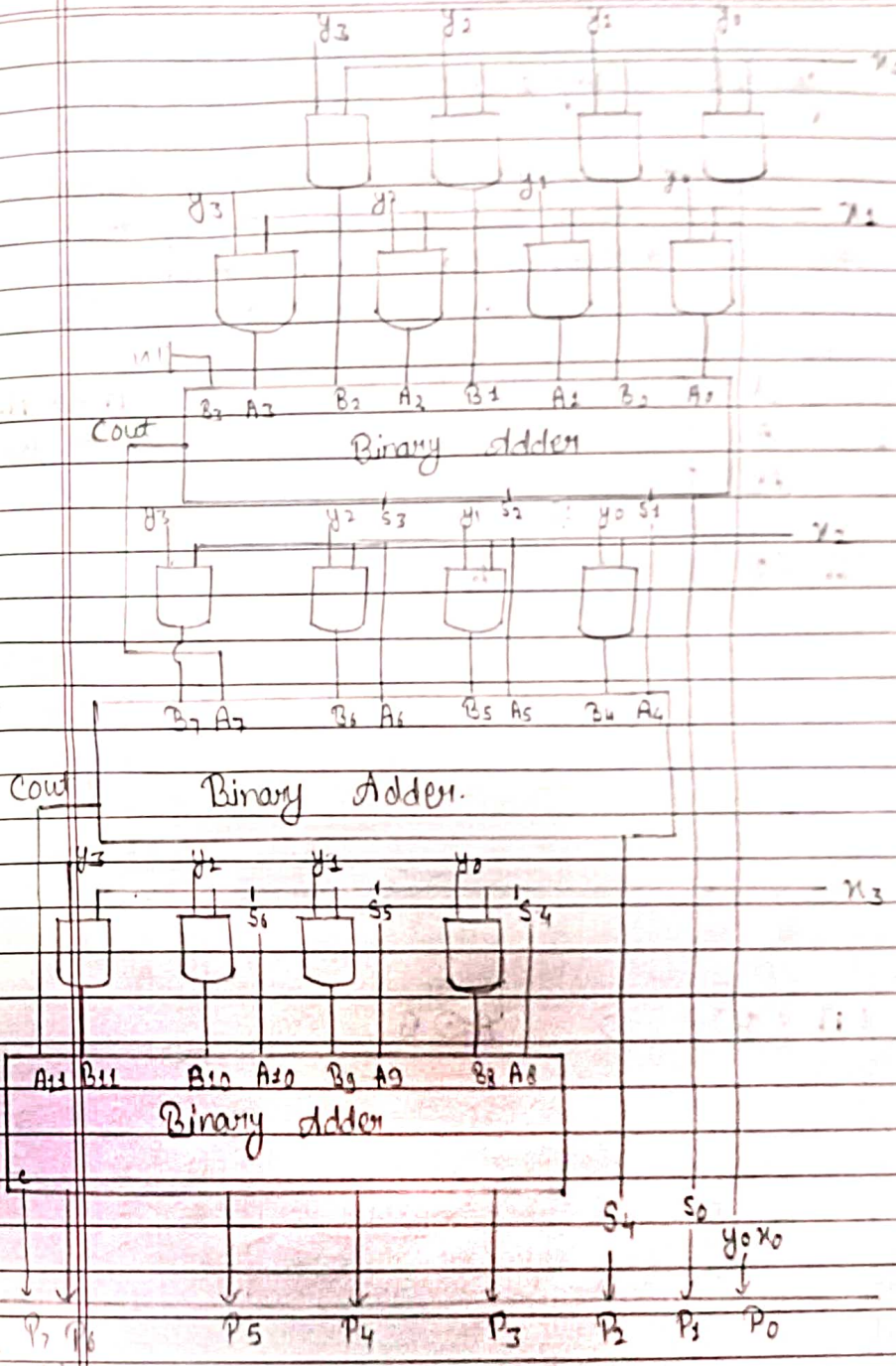
y_3x_2 y_2x_2 y_1x_2 y_0x_2 × ×

y_3x_3 y_2x_3 y_1x_3 y_0x_3 × × ×

Answer

P_6 P_5 P_4 P_3 P_2 P_1 P_0

Circuit diagram for four bit number



★ Magnitude Comparator

A magnitude comparator is a hardware electronic device that takes two numbers as input in binary form and determines whether one number is greater than, less than or equal to the other numbers.

for ~~two~~ one bits.

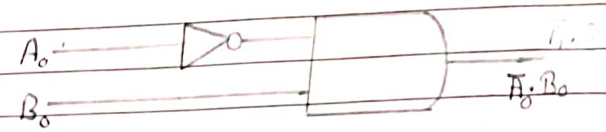
1. $A < B = 0$
2. $A = B = 1$
3. $A > B = 0$

$\{A \rightarrow A_0, B \rightarrow B_0\}$

(i) $(A < B) \Rightarrow \bar{A_0} \cdot B_0$

T.T

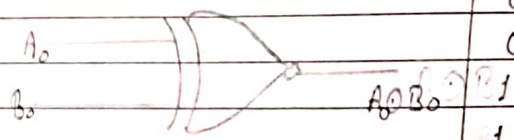
A	B	$\bar{A} \cdot B$
0	0	0
0	1	1
1	0	0
1	1	0



(ii) $(A = B) \Rightarrow A_0 \odot B_0$

T.T

A	B	$A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1



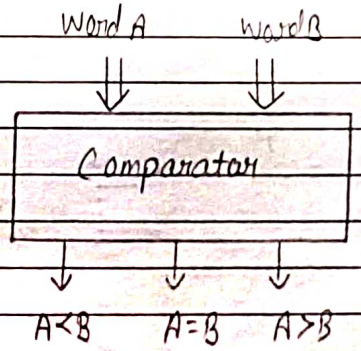
(iii) $(A > B) \Rightarrow A_0 \cdot \bar{B_0}$

TT

A	B	$A \cdot \bar{B}$
0	0	0
0	1	0
1	0	1
1	1	0

★ 2-Bit Comparator (Conventional Method)

A digital comparator is a combinational circuit designed to compare two n-bit binary words. 2-bit comparator has three outputs.



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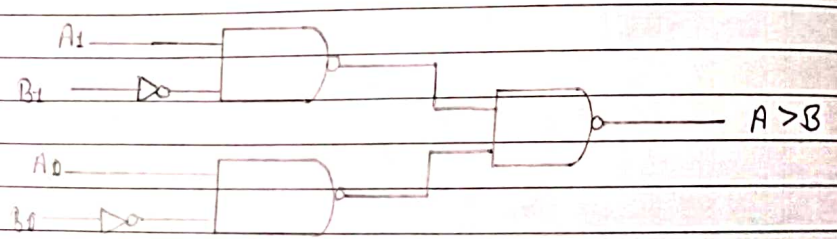
Truth Table Of Two-Bit Magnitude Comparator

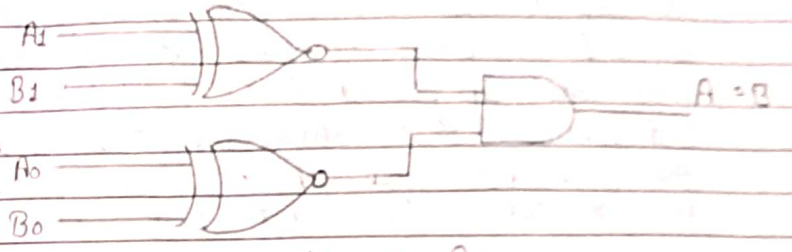
	INPUT				OUTPUT		
	A_1	A_0	B_1	B_0	A > B	A = B	A < B
0	0	0	0	0	0	1	0
1	0	0	0	1	0	0	1
2	0	0	1	0	0	0	1
3	0	0	1	1	0	0	1
4	0	1	0	0	1	0	0
5	0	1	0	1	0	1	0
6	0	1	1	0	0	0	1
7	0	1	1	1	0	0	1
8	1	0	0	0	1	0	0
9	1	0	0	1	1	0	0
10	1	0	1	0	0	1	0
11	1	0	1	1	0	0	1
12	1	1	0	0	1	0	0
13	1	1	0	1	1	0	0
14	1	1	1	0	1	0	0
15	1	1	1	1	0	1	0

(i) (A > B)



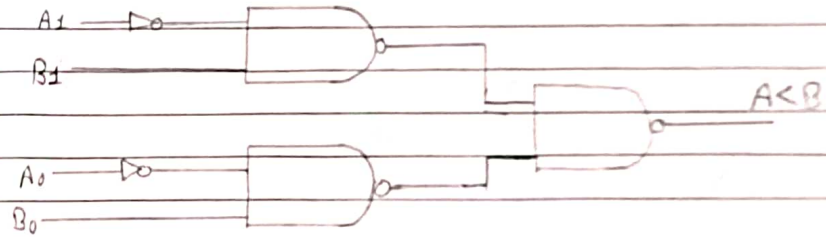
Circuit diagram

(ii) (A = B)



Circuit Diagram

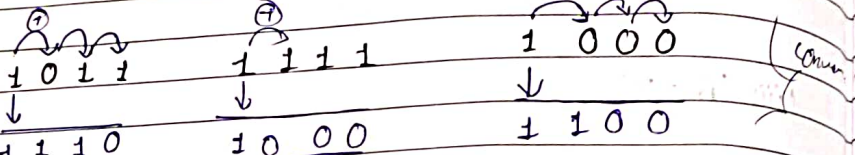
(iii) (A < B)



Circuit Diagram

☆ Binary to Gray code converter

The binary to Gray code converter is a logical circuit that is used to convert the binary code into its equivalent Gray code. By putting the MSB of 1 below the axis and the MSB of 1 above the axis and reflecting the (n-1) bit code about an axis after 2^{n-1} rows we can obtain the n-bit gray code.

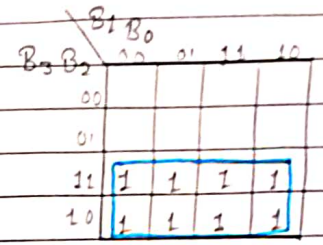


Truth Table of Binary to Gray Code Converter

	Input				Output			
	B_3	B_2	B_1	B_0	G_3	G_2	G_1	G_0
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
2	0	0	1	0	0	0	1	1
3	0	0	1	1	0	0	1	0
4	0	1	0	0	0	1	1	0
5	0	1	0	1	0	1	1	1
6	0	1	1	0	0	1	0	1
7	0	1	1	1	0	1	0	0
8	1	0	0	0	1	1	0	0
9	1	0	0	1	1	1	0	1
10	1	0	1	0	1	1	1	1
11	1	0	1	1	1	1	1	0
12	1	1	0	0	1	0	1	0
13	1	1	0	1	1	0	1	1
14	1	1	1	0	1	0	0	1
15	1	1	1	1	1	0	0	0

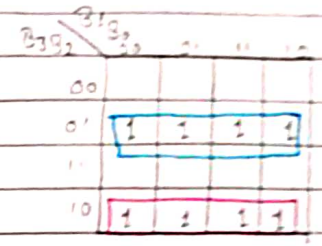
Expression of the truth table with the help of K-map

- for G_3



$G_3 = B_3$

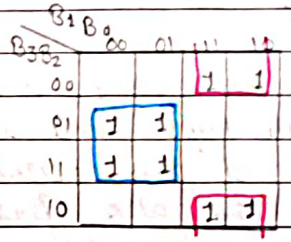
- for G_2



$G_2 = \bar{B_3}B_2 + B_3\bar{B_2}$

$\Rightarrow B_3 \oplus B_2$

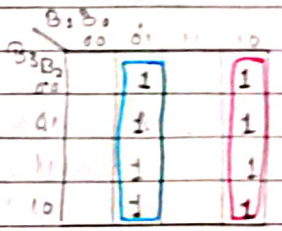
- for G_1



$G_1 = \bar{B_2}B_1 + B_2\bar{B_1}$

$\Rightarrow B_2 \oplus B_1$

- for G_0



$G_0 = \bar{B_1}B_0 + \bar{B_0}B_1$

$\Rightarrow B_1 \oplus B_0$

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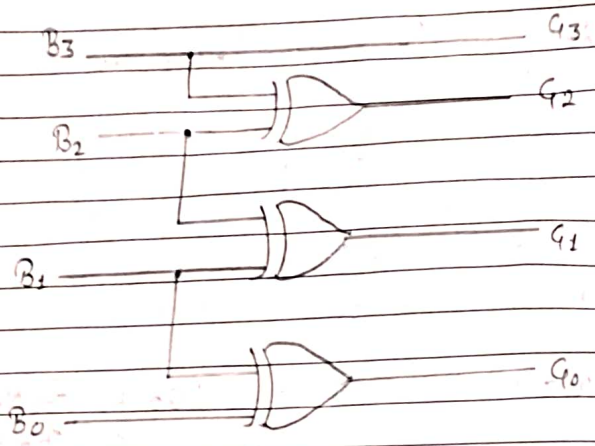
• Circuit Diagram for Binary to gray code converter



☆ Gray to Binary Code Converter

Gray codes are used in rotary and optical encoders, Karnaugh maps, and error detection. The hamming distance of two neighbours gray codes is always 1 and also first Gray code and last Gray code also has Hamming distance is always 1. So it is also called Cyclic codes.

Methods { 1. Using K-Map
2 Using EX-OR gate

1 (1) 0 1
↓ (+)
(1) 0 0 1

1 (1) 1 1
↓ (+)
(1) 0 1 0

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• Truth Table for Gray to Binary code Converter

	Input				Output			
	G_3	G_2	G_1	G_0	B_3	B_2	B_1	B_0
0.	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
2	0	0	1	0	0	0	1	1
3	0	0	1	1	0	0	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	0	1	1	0
6	0	1	1	0	0	1	0	0
7	0	1	1	1	0	1	0	0
8	1	0	0	0	1	1	1	1
9	1	0	0	1	1	1	1	0
10	1	0	1	0	1	1	0	0
11	1	0	1	1	1	1	0	0
12	1	1	0	0	1	0	0	0
13	1	1	0	1	1	0	0	1
14	1	1	1	0	1	0	1	1
15	1	1	1	1	1	0	1	0

• Expressions of the truth table with the help of K-Map

• for B_3

G_3G_2 \ G_1G_0	00	01	11	10
00				
01				
11	1	1	1	1
10	1	1	1	1

$B_3 = G_3$

• for B_2

G_3G_2 \ G_1G_0	00	01	11	10
00				
01	1	1	1	1
10				
11	1	1	1	1

$B_2 = \overline{G_3}G_2 + G_3\overline{G_2} \Rightarrow G$

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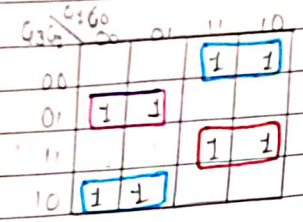
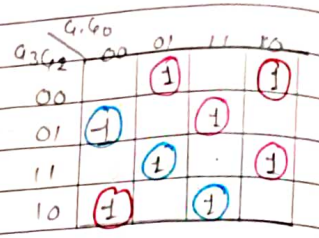
• for B_1



$B_1 \Rightarrow G_3\bar{G}_2\bar{G}_1 + \bar{G}_3G_2\bar{G}_1 + \bar{G}_3\bar{G}_2G_1 + G_3G_2G_1$

$\Rightarrow G_3 \oplus G_2 \oplus G_1$

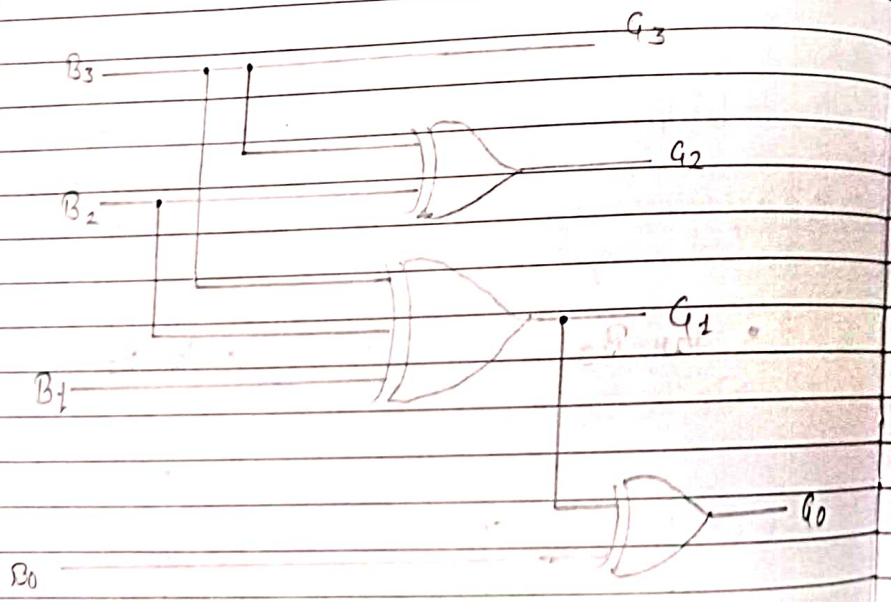
• for B_0



$B_0 \Rightarrow G_3\bar{G}_2\bar{G}_1\bar{G}_0 + \bar{G}_3G_2G_1\bar{G}_0 + \bar{G}_3G_2\bar{G}_1G_0 + \bar{G}_3\bar{G}_2G_1G_0$

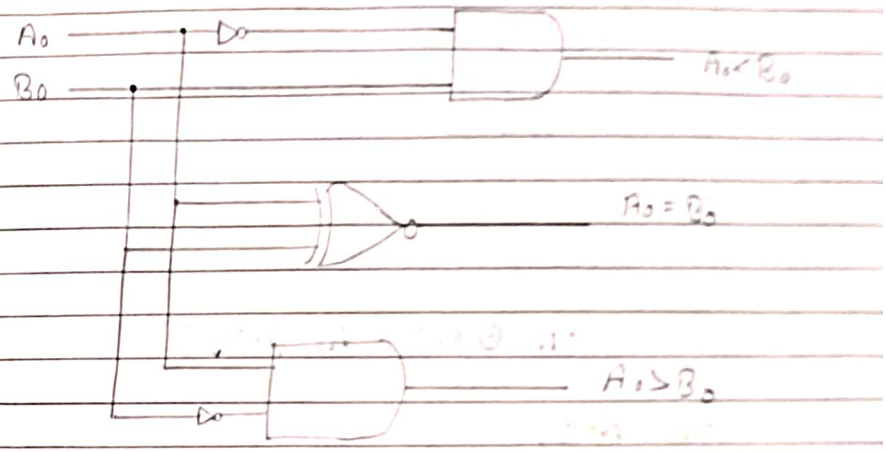
$\Rightarrow G_3 \oplus G_2 \oplus G_1 \oplus G_0$

• Circuit Diagram for gray to Binary Code converter



Magnitude Comparator :

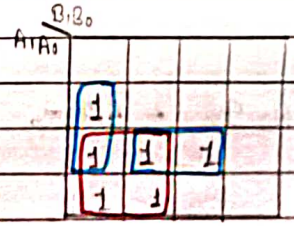
• Circuit Diagram for one Bit



• Circuit Diagram for two bit.

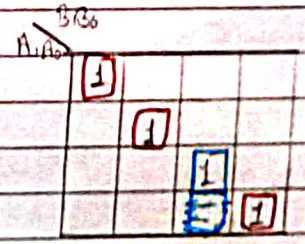
K-Map for the truth table

for $A > B$,



$\Rightarrow A_1\bar{B}_1 + A_0\bar{B}_1\bar{B}_0 + A_1A_0\bar{B}_0$

for $A = B$,



$\Rightarrow \bar{A}_1\bar{B}_1 + \bar{A}_1\bar{A}_0\bar{B}_0 + \bar{A}_0\bar{B}_1\bar{B}_0$
$\bar{A}_1\bar{A}_0\bar{B}_1\bar{B}_0 + \bar{A}_1A_0\bar{B}_1B_0 + A_1A_0B_1B_0$
$+ A_1\bar{A}_0B_1\bar{B}_0$

$\Rightarrow (A_1 \odot B_1)(A_0 \odot B_0)$

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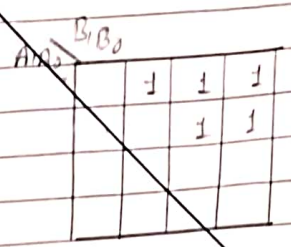
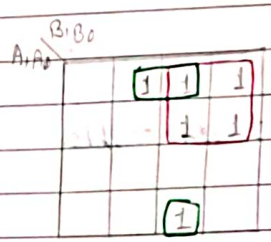
for $A < B$,



$\Rightarrow \overline{A_1}\,\overline{A_0}\,\overline{B_1}\,\overline{B_0} + \overline{A_1}\,A_0\,\overline{B_1}\,B_0 + A_1A_0B_1B_0 + A_1\overline{A_0}\,B_1\overline{B_0}$

$\Rightarrow (A_1 \odot B_1)(A_0 \odot B_0)$

for $A < B$,



$\Rightarrow \overline{A_1}B_1 + \overline{A_1}\,\overline{A_0}\,B_0 + \overline{A_0}B_1B_0$

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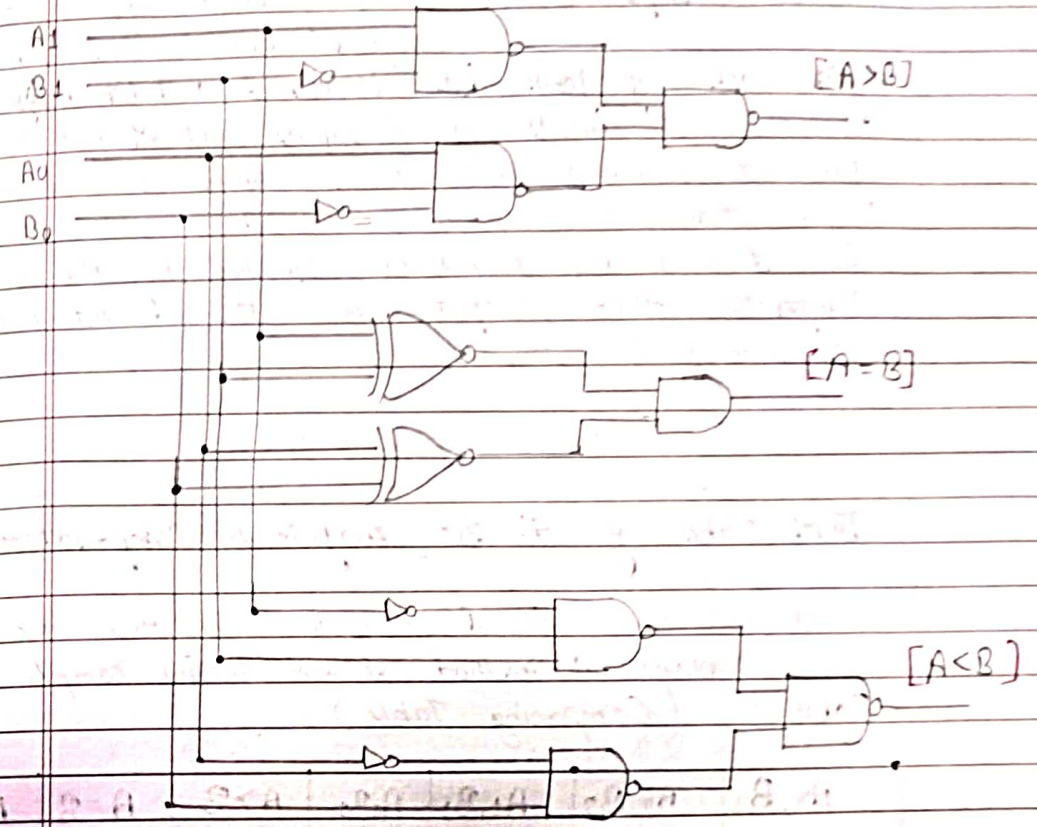
Circuit Diagram for two bit Magnitude Compa



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Magnitude Comparator

☆ **4-Bit Comparator** :- A comparator applied to Compare two binary numbers with each of four bits is termed a 4-bit magnitude comparator. It consists of 8 inputs each of for two four-bit numbers and three outputs to generate less than (<), equal to (=) and greater than (>) b/w two binary numbers. Similar to the 3-bit comparators the output for the 4-bit magnitude comparators is obtained.

Truth Table of 4-Bit magnitude Comparator :-

We will use general method for this bcz by the conventional method it will become complicated.

Table : (Comparing Table)

A_3, B_3	A_2, B_2	A_1, B_1	$A_0 B_0$	$A > B$	$A = B$	$A < B$
$A_3 > B_3$	X	X	X	1	0	0
$A_3 = B_3$	$A_2 > B_2$	X	X	1	0	0
$A_3 = B_3$	$A_2 = B_2$	$A_1 > B_1$	X	1	0	0
$A_3 = B_3$	$A_2 = B_2$	$A_1 = B_1$	$A_0 > B_0$	1	0	0
$A_3 < B_3$	X	X	X	0	0	1
$A_3 = B_3$	$A_2 < B_2$	X	X	0	0	1
$A_3 = B_3$	$A_2 = B_2$	$A_1 < B_1$	X	0	0	1
$A_3 = B_3$	$A_2 = B_2$	$A_1 = B_1$	$A_0 < B_0$	0	0	1
$A_3 = B_3$	$A_2 = B_2$	$A_1 = B_1$	$A_0 = B_0$	0	1	0
$A_3 \odot B_3 = X_3$	$A_2 \odot B_2 = X_2$	$A_1 \odot B_1 = X_1$	$A_0 \odot B_0 = X_0$			

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Next Table. (Truth Table)

A	B	A > B	A = B	A < B
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

$A = B \Rightarrow (A_3 \odot B_3) . (A_2 \odot B_2) . (A_1 \odot B_1) . (A_0 \odot B$

$A > B \Rightarrow A_3 \overline{B_3} + X_3 (A_2 \overline{B_2}) + X_3 . X_2 . (A_1 \overline{B_1}) + X_3 X_2 X_1 (A_0 \overline{B_0})$

$A < B \Rightarrow \overline{A_3} B_3 + X_3 (\overline{A_2} B_2) + X_3 X_2 (\overline{A_1} B_1) + X_3 X_2 X_1 (\overline{A_0} B_0)$

- $A > B \Rightarrow A \overline{B}$
- $A = B \Rightarrow \overline{A}\,\overline{B} + AB = A \odot B$
- $A < B \Rightarrow \overline{A} B$

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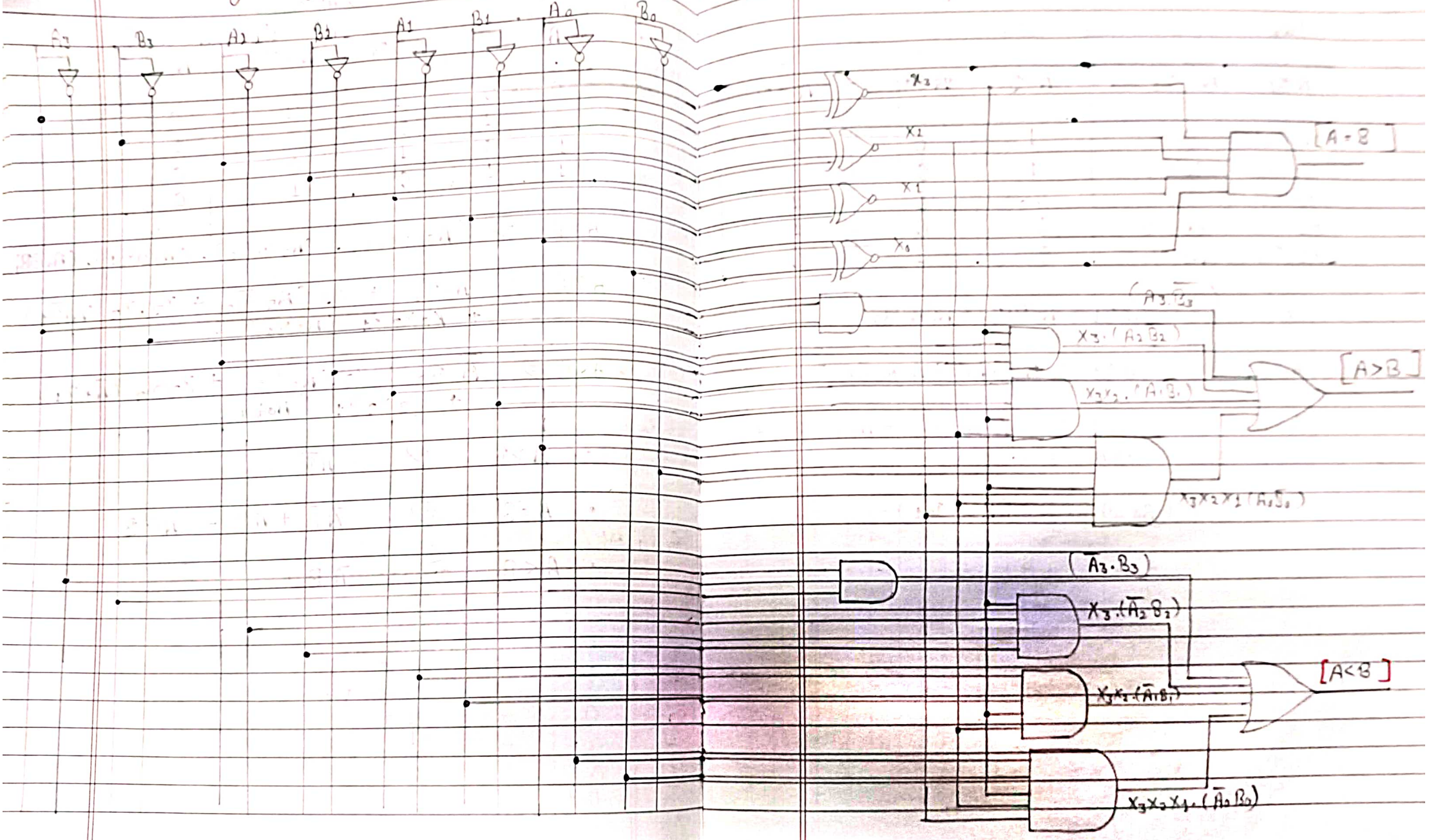
Circuit Diagram for 4-Bit Magnitude Comparator



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★ 2-Bit Magnitude comparator using General Method

Table :- (Comparing Table)

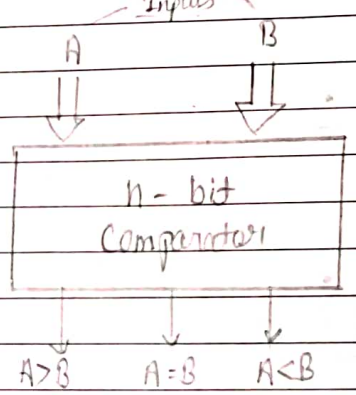
A_1B_1	A_0, B_0	$A > B$	$A = B$	$A < B$
$A_1 > B_1$	X	1	0	0
$A_1 = B_1$	$A_0 > B_0$	1	0	0
$A_1 < B_1$	X	0	0	1
$A_1 = B_1$	$A_0 < B_0$	0	0	1
$A_1 = B_1$	$A_0 = B_0$	0	1	0

$(A<B) \Rightarrow A_1'B_1 + A_0'B_1B_0 + A_1'A_0'B_0$

$(A>B) \Rightarrow A_1B_1' + A_0B_1'B_0' + A_1A_0B_0$

$(A=B) \Rightarrow A_1'A_0'B_1'B_0' + A_1'A_0B_1'B_0 + A_1A_0B_1B_0 + A_1A_0'B_1B_0'$

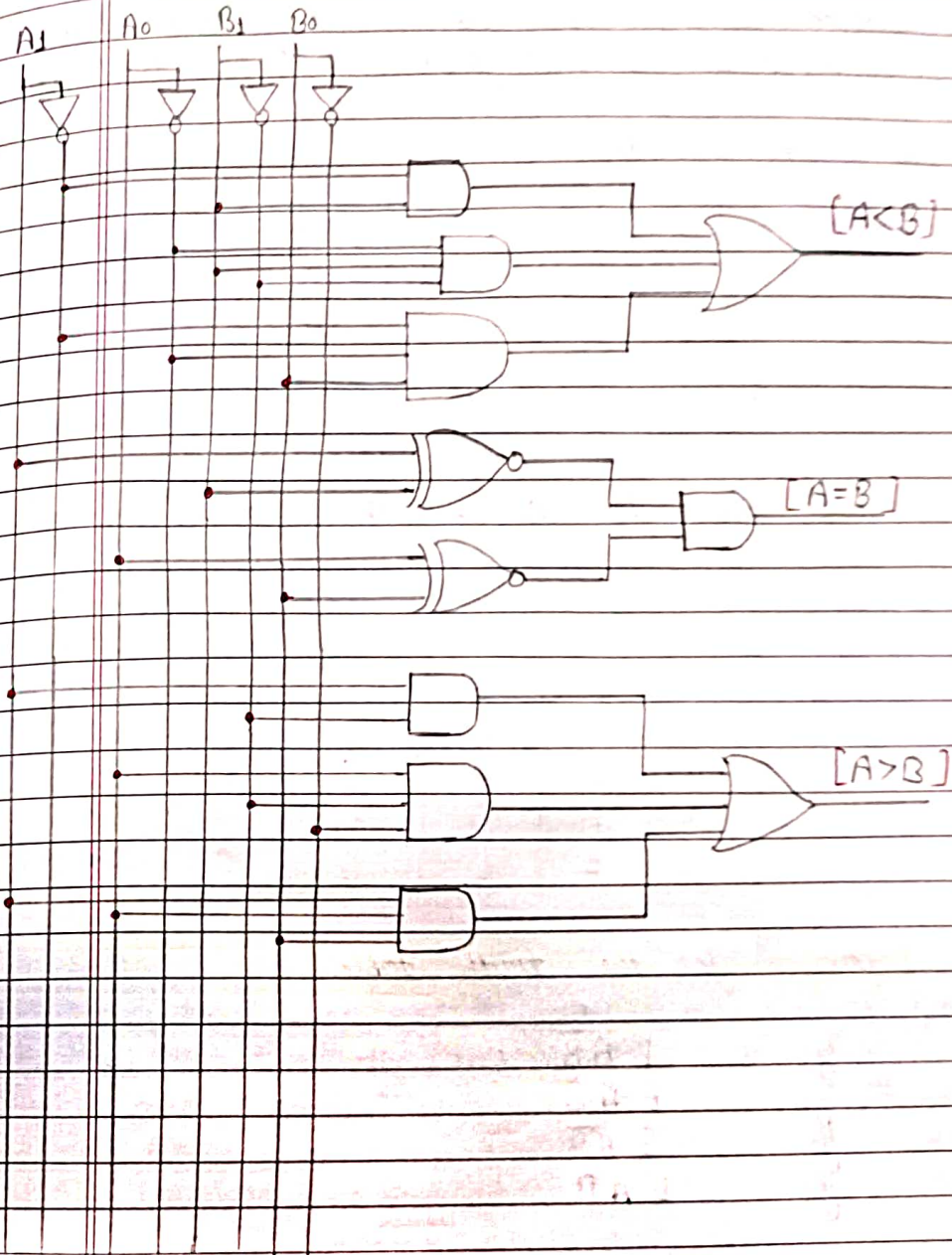
$\Rightarrow (A_0 \odot B_0)\ (A_1 \odot B_1)$



⇒ Comparator

n - bit

Logic Diagram of 2 bit Comparator



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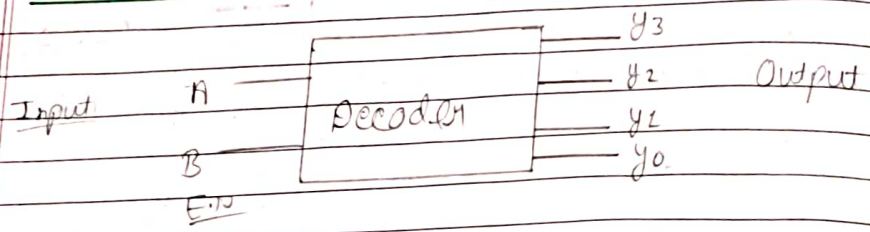
Decoder

Decoder is a combinational circuit which has n input lines will give 2^N output lines.

2 to 4 Decoder



Truth Table for the 2 to 4 Decoder.

EN	Inputs		Outputs			
	A	B	y_3	y_2	y_1	y_0
0	X	X	X	X	X	X
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

Expression for the truth table.

$y_0 = E\bar{A}\bar{B}$
$y_1 = E\bar{A}B$
$y_2 = EA\bar{B}$
$y_3 = EAB$

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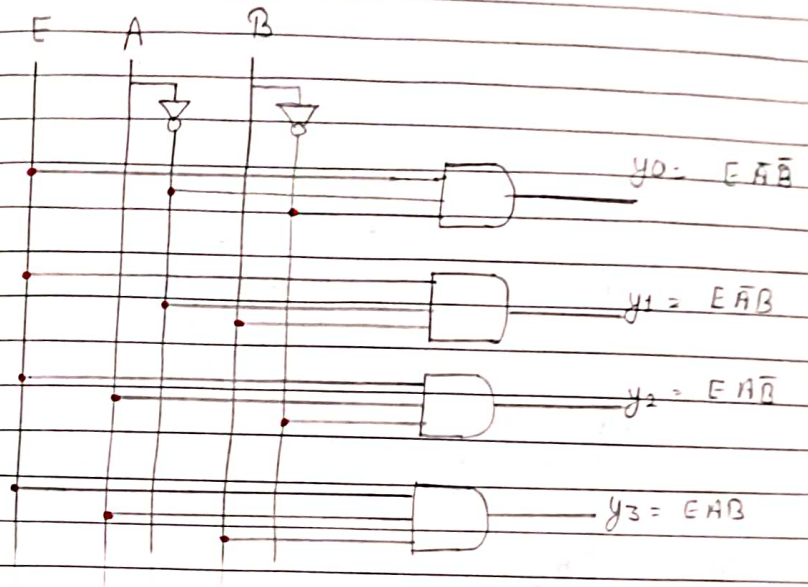
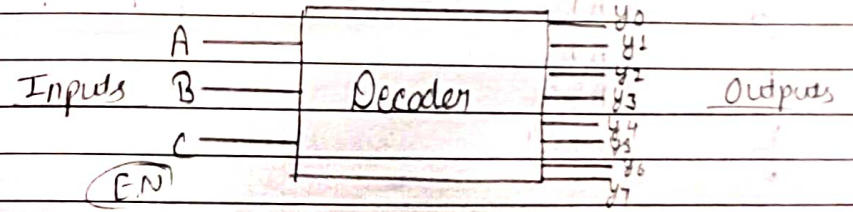
Circuit Diagram for 2 to 4 Decoder



3 to 8 Decoder



3 to 8 line decoder demultiplexer is a combinational circuit that can be used as both a decoder and a demultiplexer. IC 74HC238 decodes three binary address inputs (A0, A1, A2) into eight outputs (Y0 to Y7). The device also has three enable pins.

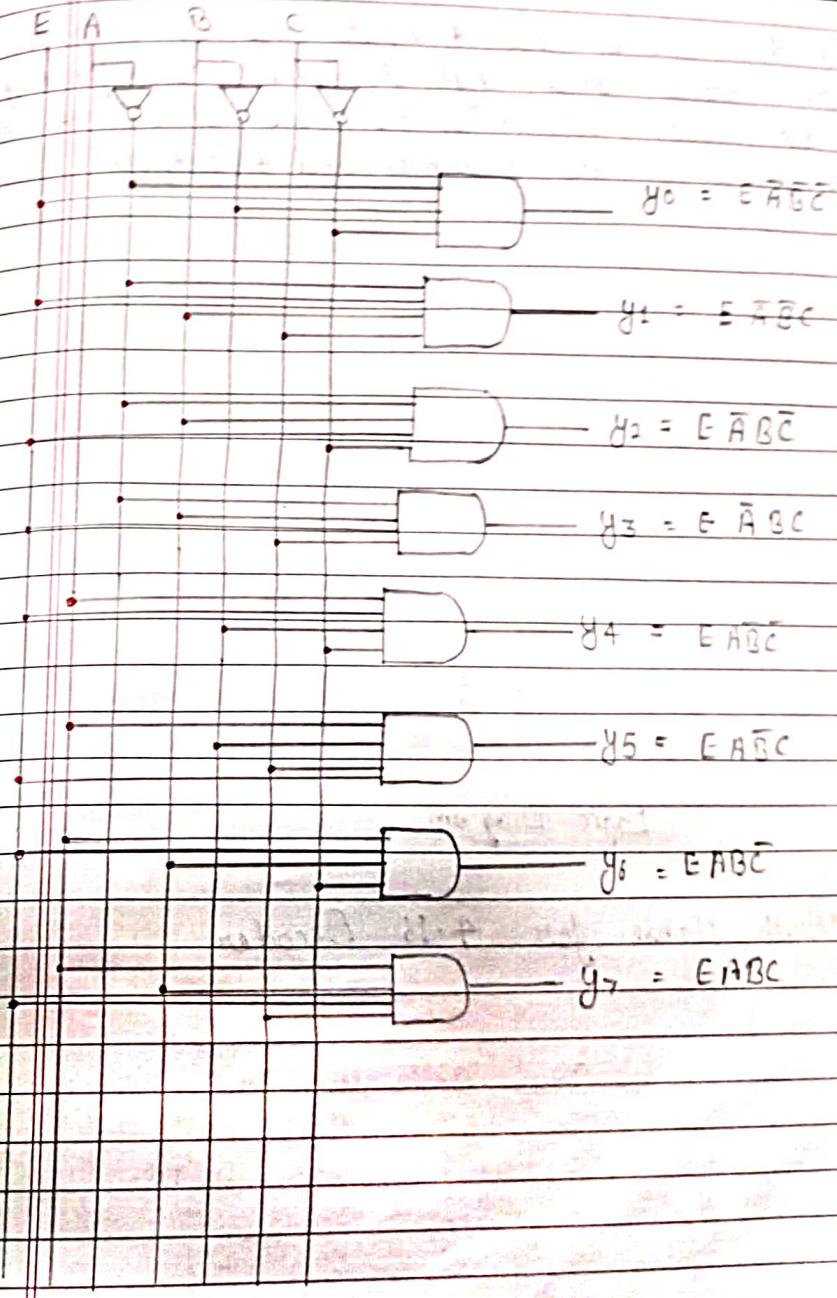
Truth Table for the 3 to 8 Decoder

EN	Inputs			Outputs							
	A	B	C	y_7	y_6	y_5	y_4	y_3	y_2	y_1	y_0
0	x	x	x	x	x	x	x	x	x	x	x
1	0	0	0	0	0	0	0	0	0	0	1
1	0	0	1	0	0	0	0	0	0	1	0
1	0	1	0	0	0	0	0	0	1	0	0
1	0	1	1	0	0	0	0	1	0	0	0
1	1	0	0	0	0	0	1	0	0	0	0
1	1	0	1	0	0	1	0	0	0	0	0
1	1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	1	0	0	0	0	0	0	0

$y_0 = E\bar{A}\bar{B}\bar{C}$
$y_1 = E\bar{A}\bar{B}C$
$y_2 = E\bar{A}B\bar{C}$
$y_3 = E\bar{A}BC$
$y_4 = EA\bar{B}\bar{C}$
$y_5 = EA\bar{B}C$
$y_6 = EAB\bar{C}$
$y_7 = EABC$

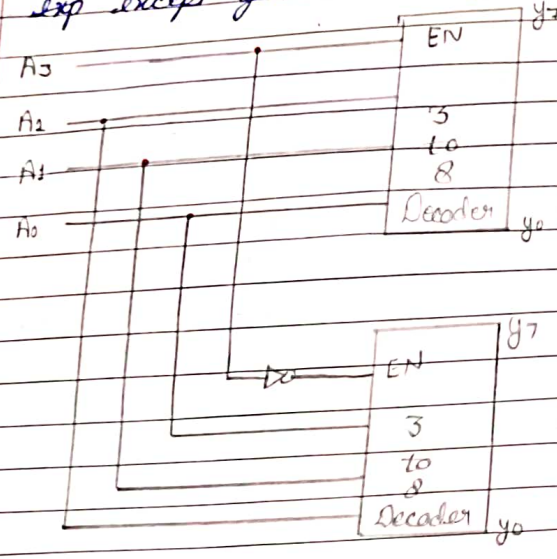
* A 3 to 8 decoder has three inputs (A, B, C) and eight outputs (y_0 to y_7). Based on the 3 inputs one of the eight outputs is selected.

Circuit Diagram for 3 to 8 Decoder



★ Design a 4 to 16 Decoder using 3 to 8 Decoder

A 4-16 decoder has 4 inputs and 16 outputs. With the outputs going high for the corresponding 4-bit input. Similar is the case of a 2×4 decoder except for its 2 inputs and 4 outputs.



Logic Diagram

Truth Table for 4-16 Decoder

Input	Output			
	A_3	A_2	A_1	A_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

★ Implementation of higher order decoders using lower order decoders

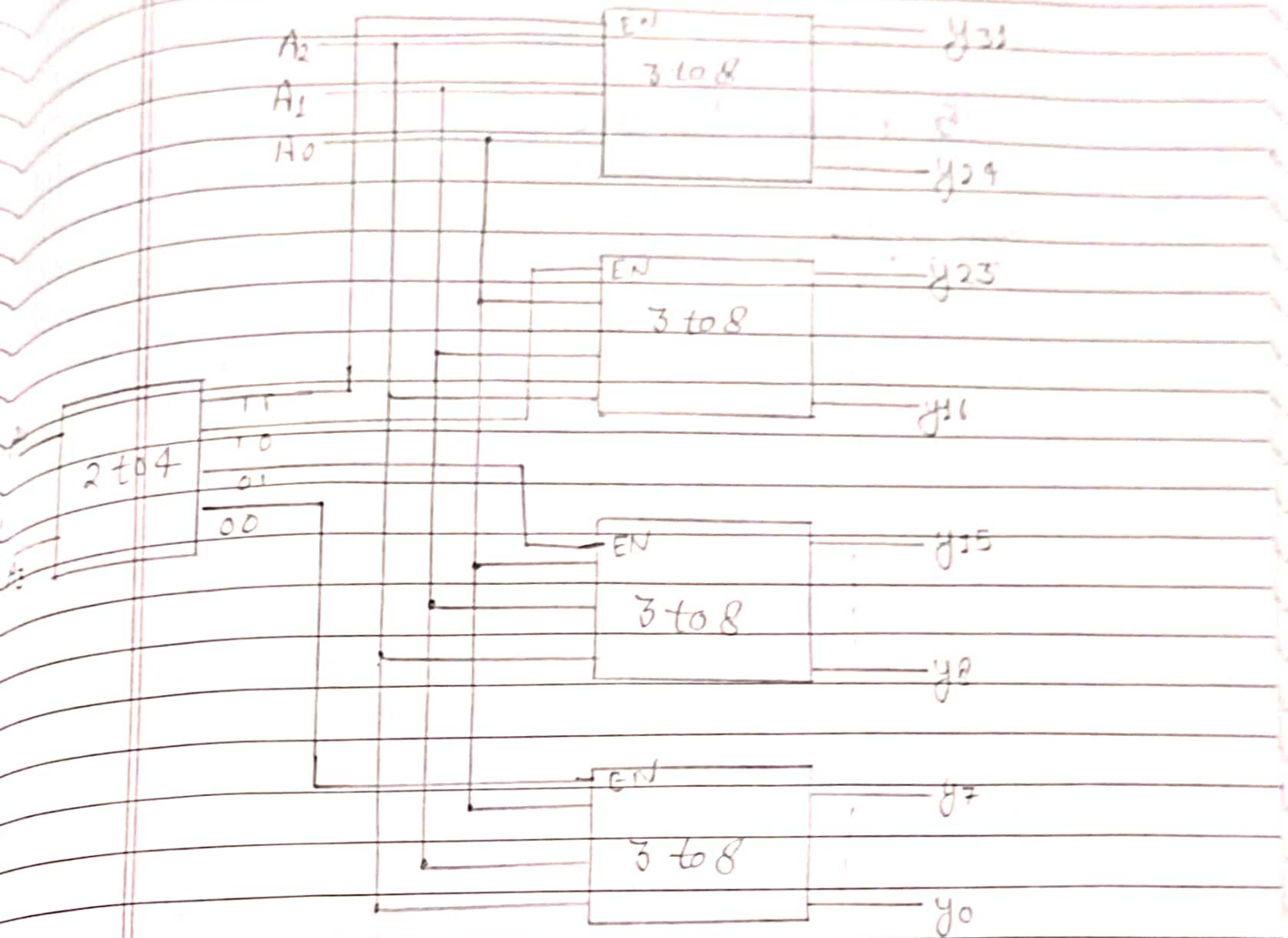
- 5 to 32 using 3 to 8 Decode.

$$\frac{32}{8} = 4$$

[4 ⇒ 3 to 8 Decoder]

⇒ 5 to 32 decoder means that only one output will be active out of the 32 output. First Step - you need 32 outputs so use four 3 to 8 decoders. Each one will use the 3 LSB controls of the 5 input signals. Since they will output 4 active outputs you will need to control the enable pin so only 1 will be enabled.

Logic Diagram :-

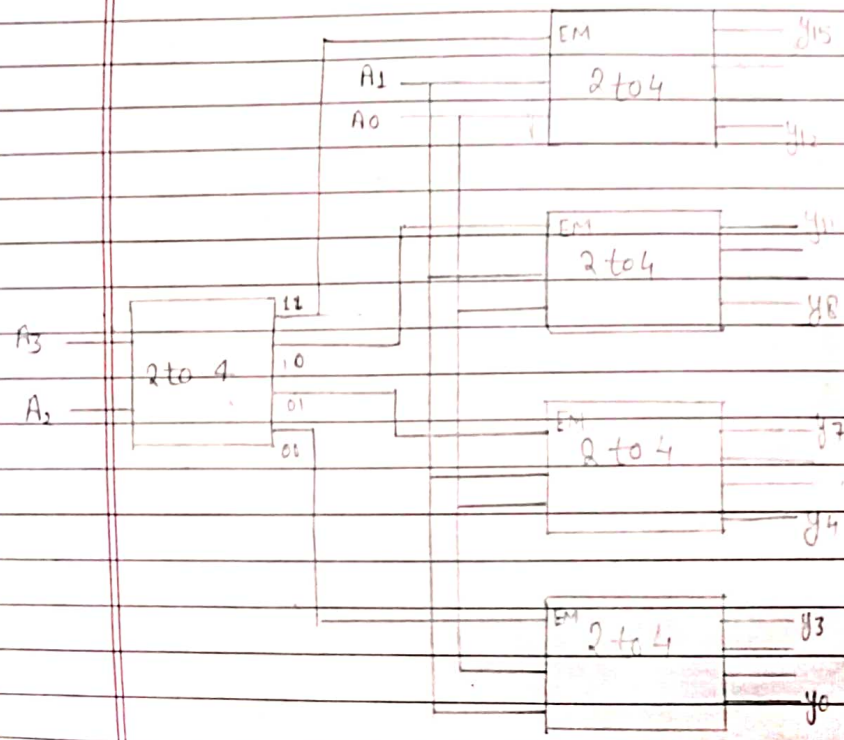


- 4 to 16 Decoder using 2 to 4

$\frac{16}{4} = 4$, $\frac{4}{4} = 1$; $4+1 = 5$

5 → [2 to 4 Decoder]

Logic Diagram



- Applications of Decoder

★ BCD to 7 segment Decoder

In Binary Coded Decimal (BCD) encoding scheme each of the decimal numbers (0-9) is represented by its equivalent binary pattern (which is generally of 4 bits).

Truth Table

	Input				Output (BCD)						
	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1

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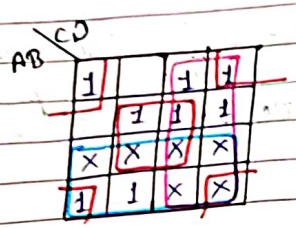
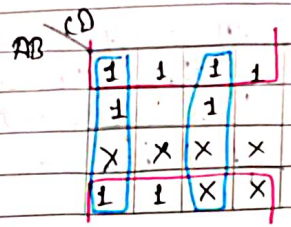
K Map :-

1. for a



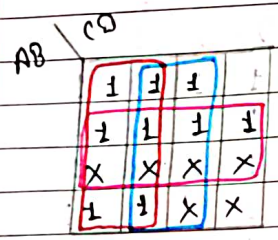
$a \Rightarrow C + A + B\bar{D} + \bar{B}\bar{D}$

2. for b



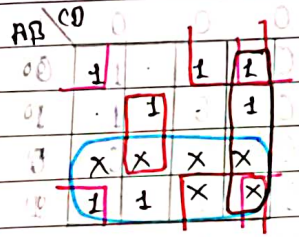
$b \Rightarrow \bar{B} + \bar{C}\bar{D} + CD$

3. for c



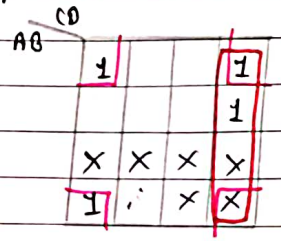
$c \Rightarrow B + \bar{C} + D$

4. for d



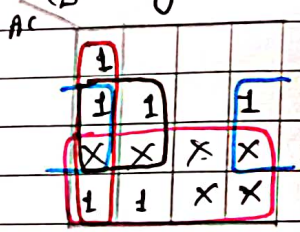
$d \Rightarrow A + C\bar{D} + \bar{B}\bar{D} + \bar{B}C + B\bar{C}D$

5. for e



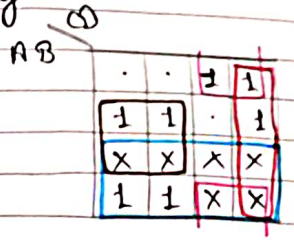
$e \Rightarrow C\bar{D} + \bar{B}\bar{D}$

6. for f



$f \Rightarrow A + \bar{C}\bar{D} + B\bar{C} + B\bar{D}$

7. for g



$g \Rightarrow A + C\bar{D} + B \oplus C$

Expression

$a = C + A + B\bar{D} + \bar{B}\bar{D}$

$b = \bar{B} + \bar{C}\bar{D} + CD$

$c = B + \bar{C} + D$

$d = A + C\bar{D} + \bar{B}\bar{D} + \bar{B}C + B\bar{C}D$

$e = C\bar{D} + \bar{B}\bar{D}$

$f = A + \bar{C}\bar{D} + B\bar{C} + B\bar{D}$

$g = A + C\bar{D} + B \oplus C$

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Circuit Diagram :-

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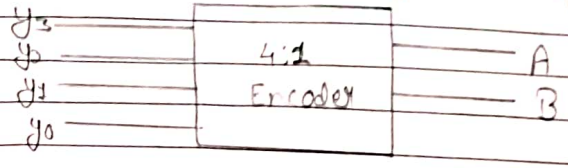
Encoders :-

An encoder in digital electronics is a one-hot to binary converter. That is, if there are 2^n input lines, and at most only one of them will ever be high, the binary code of this 'hot' line is produced on the n-bit output lines.

Block Diagram

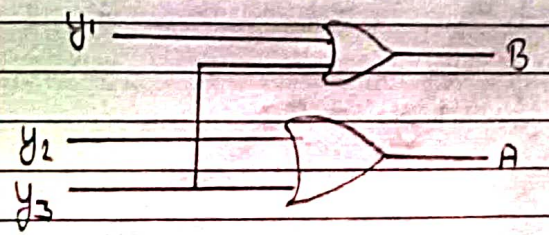


Truth Table [4:2]

E	Input				Output	
	y_3	y_2	y_1	y_0	A	B
0	X	X	X	X	X	X
1	0	0	0	1	0	0
1	0	0	1	0	0	1
1	0	1	0	0	1	0
1	1	0	0	0	1	1

Decoder	Encoders
1:2	2:1
2:4	4:2
3:8	8:3
4:16	16:4
.	.
$n = 2^n$	$2^n = n$

$$A = y_2 + y_3$$
$$B = y_1 + y_3$$



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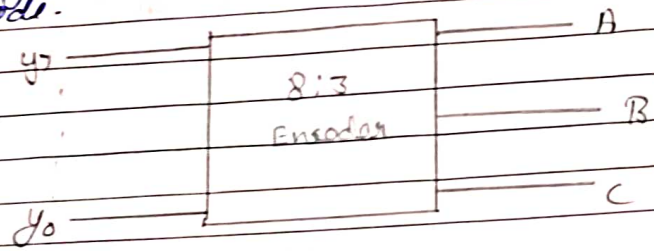
★ [8:3] Encoder

The 8 to 3 Encoder or octal to Binary encoder consists of 8 inputs : Y_7 to Y_0 and 3 outputs A, B & C. Each input line corriesponds to each octal digit and three outputs generate corresponding binary code.



Truth Table

	E	Input								Output		
		y_7	y_6	y_5	y_4	y_3	y_2	y_1	y_0	A	B	C
	0	×	×	×	×	×	×	×	×	×	×	×
0	1	0	0	0	0	0	0	0	1	0	0	0
1	1	0	0	0	0	0	0	1	0	0	0	1
2	1	0	0	0	0	0	1	0	0	0	1	0
3	1	0	0	0	0	1	0	0	0	0	1	1
4	1	0	0	0	1	0	0	0	0	1	0	0
5	1	0	0	1	0	0	0	0	0	1	0	1
6	1	0	1	0	0	0	0	0	0	1	1	0
7	1	1	0	0	0	0	0	0	0	1	1	1

$A = y_4 + y_5 + y_6 + y_7$

$B =$

$C =$

Decoder : $n \rightarrow 2^n$
encoder : $2^n \rightarrow n$
Mux : $2^n \rightarrow 1$

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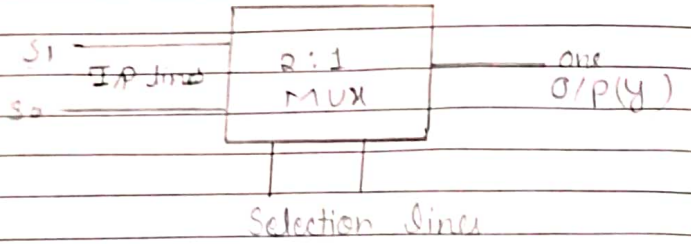
Multiplexer (Mux)

(Many to One) [Data Selector]

In electronics, a multiplexer also known as a data selector, is a device that selects b/w several analog or digital input signals and forwards the selected input to a single output line.

(2:1) Mux



Data Selector

Truth Table

E	S_1	S_0	O/P (Y)
0	X	X	X
1	0	0	y_0
1	0	1	y_1
1	1	0	y_2
1	1	1	y_3

2:1 ⇒ I/P ⇒ 2 , Sele ⇒ 2

4:1 ⇒ I/P ⇒ 4 , sele ⇒ 2

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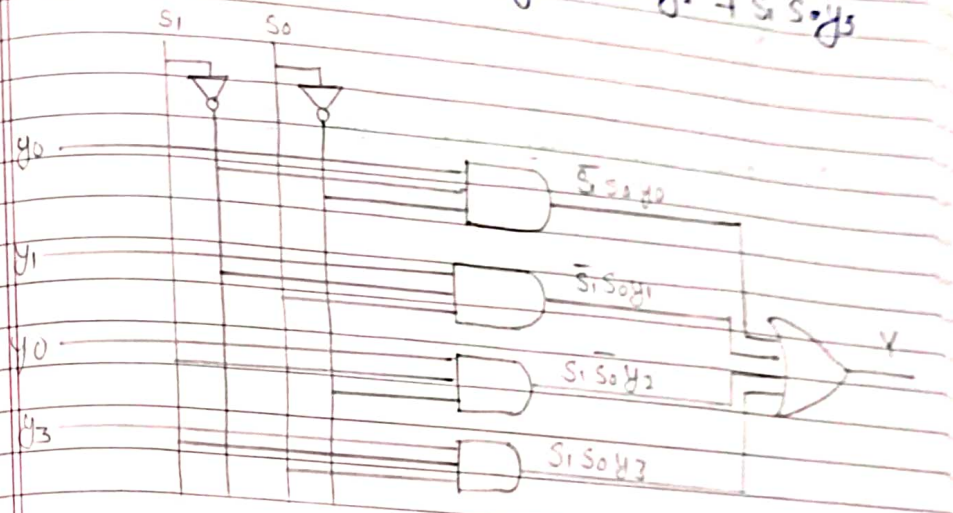
$Y = \bar{S}_1 \bar{S}_0 y_0 + \bar{S}_1 S_0 y_1 + S_1 \bar{S}_0 y_2 + S_1 S_0 y_3$



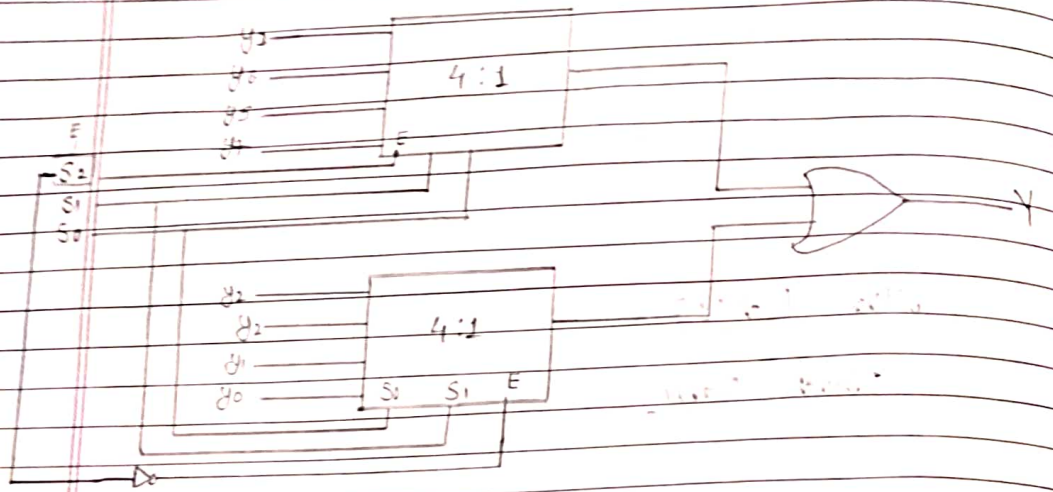
Mux [8:1]

Truth Table

E	S_2	S_1	S_0	O/P (Y)
0	X	X	X	X
1	0	0	0	y_0
1	0	0	1	y_1
1	0	1	0	y_2
1	0	1	1	y_3
1	1	0	0	y_4
1	1	0	1	y_5
1	1	1	0	y_6
1	1	1	1	y_7

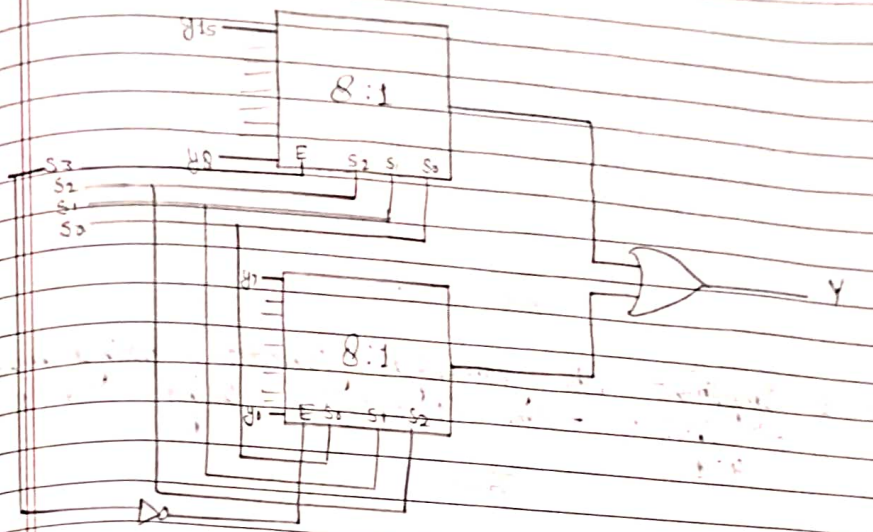
$$Y = \bar{S}_2 \bar{S}_1 \bar{S}_0 y_0 + \bar{S}_2 \bar{S}_1 S_0 y_1 + \bar{S}_2 S_1 \bar{S}_0 y_2 + \bar{S}_2 S_1 S_0 y_3 + S_2 \bar{S}_1 \bar{S}_0 y_4 + S_2 \bar{S}_1 S_0 y_5 + S_2 S_1 \bar{S}_0 y_6 + S_2 S_1 S_0 y_7$$

- **8:1 Mux with the help of 4:1 ÷**



- **16:1 Mux with the help of 8:1 ÷**

We require two 8:1 multiplexers in first stage in order to get the 16 data inputs. Since each 8:1 multiplexer produces one output we require a 2×1 Multiplexer in second stage by considering the outputs of first stage as inputs and to produce the final output.



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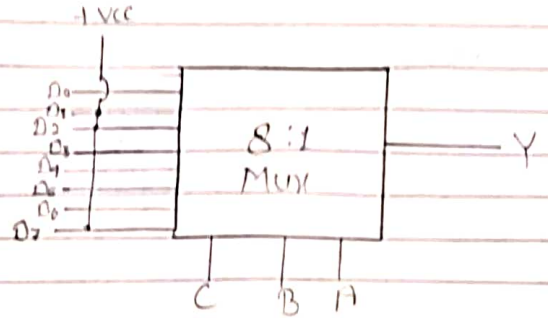
Ques. 1. Generate the logic function given in table using 8:1 mux.

Solution.

	Input			Output
	C	B	A	Y
D_0	0	0	0	0
D_1	0	0	1	1
D_2	0	1	0	1
D_3	0	1	1	0
D_4	1	0	0	0
D_5	1	0	1	0
D_6	1	1	0	0
D_7	1	1	1	1



Data Selector

Ques. 2. Implement the following function using multiplexer $f''(A,B,C) = \Sigma m\ (1,3,5,6)$ using 8:1 and 4:1.

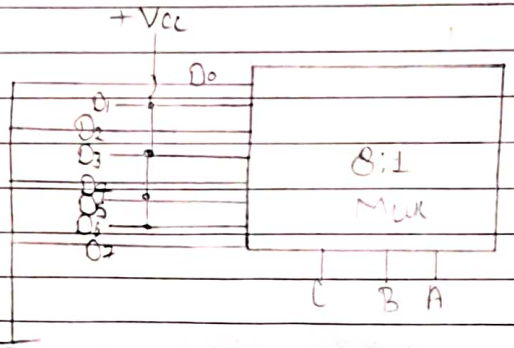
Truth Table

Solution

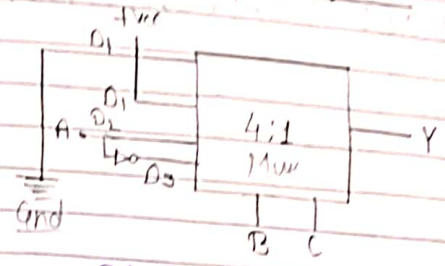
	Input			Output
	C	B	A	Y
D_1	0	0	1	1
D_3	0	1	1	1
D_5	1	0	1	1
D_6	1	1	0	1

Data Selector

8:1



4:1 Selection lines A & B

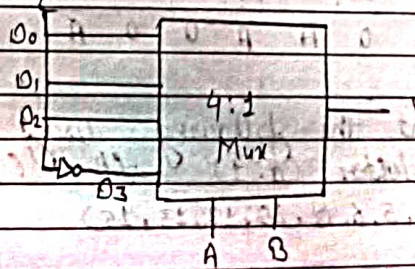


K-Map

	BC 00	BC 01	BC 10	BC 11
	D_0	D_1	D_2	D_3
$\bar{A}$	000 0	001 1	010 2	011 3
A	100 4	101 5	110 6	111 7

0 1 A $\bar{A}$

4:1 Selection lines B



K-Map

	00 D_0 AB	01 D_1 AB	10 D_2 AB	11 D_3 AB
$\bar{C}$	0	2	4	6
C	1	3	5	7

C C C $\bar{C}$

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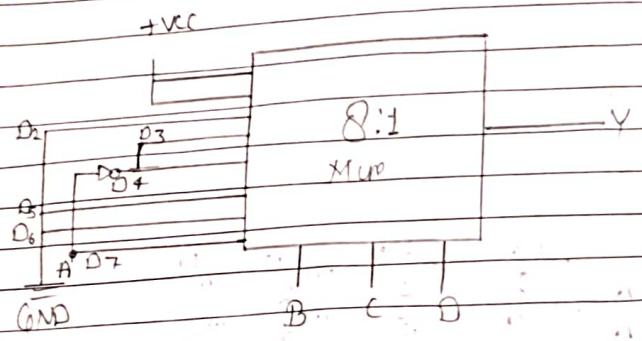
Ques :- 03. Impliment the following function using 8:1 mux
$f^n(A,B,C,D) = \Sigma m(0,1,3,4,8,9,15)$

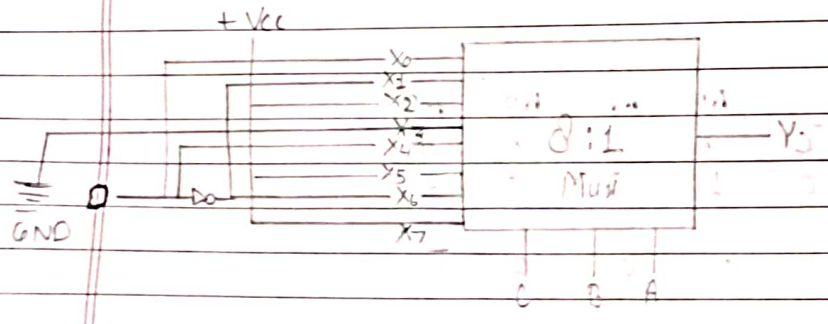
Solution.



K-Map

	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
$\bar{A}$	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15
	1	1	0	$\bar{A}$	$\bar{A}$	0	0	A

Ques :- 04 Impliment the following using 8 inputs data selector (8:1) $f^n(DCBA)$:
Σm (1,2,5,6,8,10,12,13,15)



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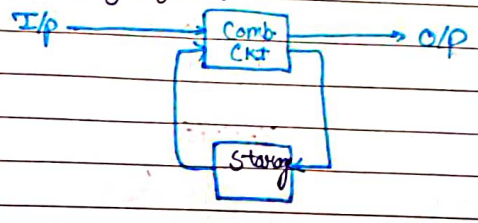
K-Map

	X_0	X_1	X_2	X_3	X_4	X_5	X_6	X_7
$\bar{D}$	0	1	2	3	4	5	6	7
D	8	9	10	11	12	13	14	15
	D	$\bar{D}$	1	0	D	1	$\bar{D}$	1

Sequential Circuits

Unit :- 03

- **Latch** ⇒ latch are digital circuit that store a single bit of information and hold its value until it is updated by new input signals
- **Flip Flop** ↳ ff are the basic piece of sequential logic. They store a single binary digit of state.



Terms use in truth table

I/p	Q_n	Q_{n+1}

NOR (Truth Table)

A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

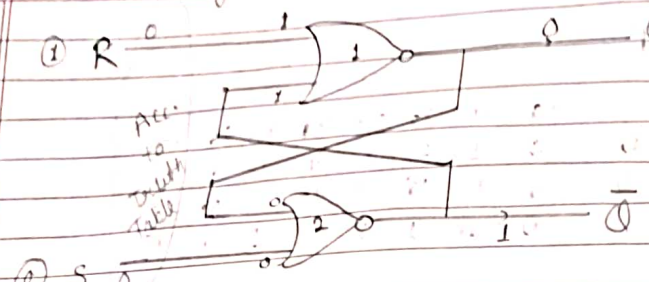
NOR Based

- S ⇒ Set
- R ⇒ Reset

★ SR latch (NOR Based)

It will store 1 I/P only

- Circuit Diagram



- Truth Table

I/P		O/P	
S	R	Q_{n+1}	$\bar{Q}_{n+1}$
0	0	Q_n	$\bar{Q}_n$ → (No change)
0	1	0	1
1	0	1	0
1	1	Indefinite / forbidden / Invalid	

Normal

for (0,0) & (1,1)

★ Assume case (1): for (0,0) condn.

- $Q = 0$
- $\bar{Q} = 1$

⇒ O/P = [0,1]

No change

Assume case (2) for (0,0) Condn.

- $Q = 1$
- $\bar{Q} = 0$

⇒ O/P = [1,0]

★ SR latch (NAND Based)

★ Flip Flop :-

(Triggering Mechanism)

Flip Flop is a sequential circuit which generally samples its inputs and changes its outputs only at particular instants of time and not continuously. Flip-flop is said to be edge sensitive or edge triggered rather than being level triggered like latches.

Clock :-

① → Level — +ve, -ve

② → Edge — +ve, -ve (Triggering) ⇒ At Transition Time.

I/P & O/P Wave forms :- (+ve) (-ve)

Assume case (1) for (1,1) Condn.

- $Q = 0$
- $\bar{Q} = 1$

⇒ O/P (0,0) ↓ Invalid

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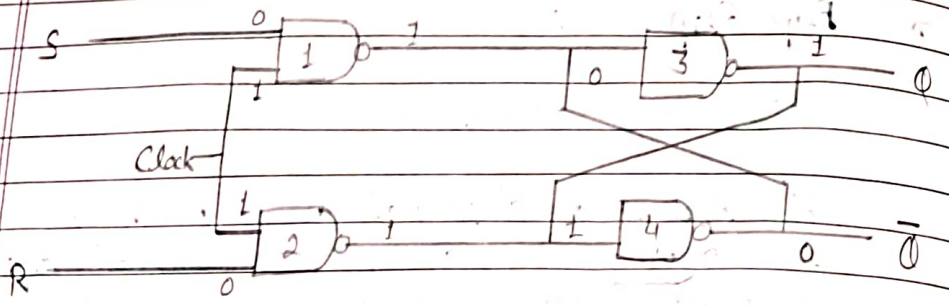
★ SR flip flop

Clock Signal :-



Truth Table :-

Clock	S	R	Q_{n+1}	$\overline{Q_{n+1}}$
0	X	X	Q_n	$\overline{Q_n}$
1	0	0	Q_n	$\overline{Q_n}$
1	0	1	0	1
1	1	0	1	0
1	1	1	Invalid	Invalid

No change

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NAND Truth Table –

A	B	$\overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

Imp

★ Present State – Next State Table

Q_n	S	R	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	Invalid
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	Invalid

‼ It is invalid for (1) so a next table made as named D flip-flop table.

★ **D Flip-flop** :- [Data flip flop] / Delay flip flop

★ Clock will always be 1 only

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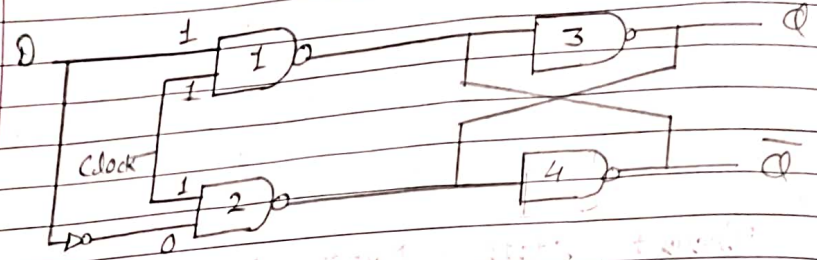
Circuit Diagram —



Truth Table :- 'Excitation table'

Clock	D	Q_{n+1}	$\overline{Q}_{n+1}$
0	X	Q_n	$\overline{Q}_n$
1	0	0	1
1	1	1	0

Only consider this condition for (0,1) & (1,0) case

★ SR Flip-Flop :-

It is a Flip Flop with two inputs, one is S and other is R. S here stands for set and R here stands for Reset. Set basically indicates set the flip flop. which means output 1 and reset indicates resetting the flip flop which mean output 1 and reset indicates resetting the flip flop which means output 0. Here Clock pluse is supplied to operate this flip flop hence it is Clocked flip flop.

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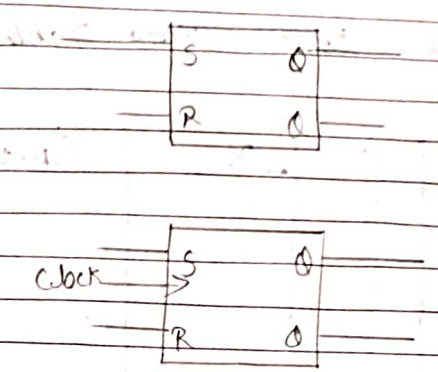
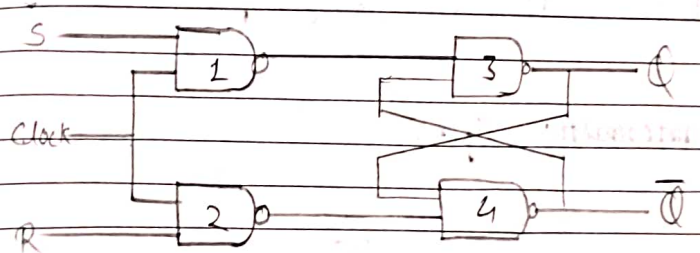


Circuit Diagram



Truth Table :-

Clock	S	R	Q_{n+1}	$\overline{Q}_{n+1}$
0	X	X	Q_n	$\overline{Q}_n$
1	0	0	Q_n	$\overline{Q}_n$
1	0	1	0	1
1	1	0	1	0
1	1	1	Invalid	

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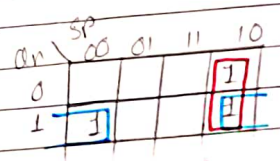
Present State, Next State Table

Q_n	S	R	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	Invalid
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	Invalid

Characteristic Eqn.



$\Rightarrow S\bar{R} + Q_n\bar{R}$

$Q_{n+1} = (S + Q_n)\bar{R}$

Excitation Table

Excitation Ipps			
Q_n	Q_{n+1}	S	R
0	0	0	d
0	1	1	0
1	0	0	1
1	1	d	0

★ ... output will not depends on prev. state

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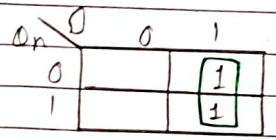
D flip-flop :-

Present State Next State Table

Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

Characteristic Eqn



$\{Q_{n+1} = D\}$

Excitation Table

Excitation Ipps		O/P
Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

⇒ 0 will operate

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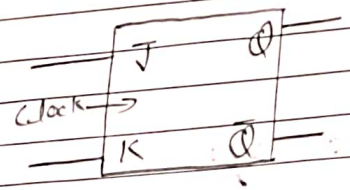
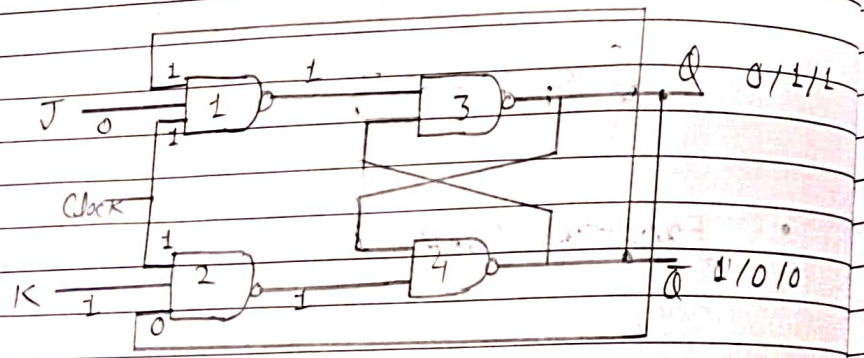
JK Flip-Flop [Jack Kilby]



- Circuit Diagram :



for 0 condition

- Truth Table [NAND]

J	K	J.K̄
0	0	1
0	1	1
1	0	1
1	1	0

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- Truth Table for JK Flip-flop

CLK	J	K	Q_{n+1}	$\bar{Q}_{n+1}$
0	X	X	Q_n	$\bar{Q}_n$
1	0	0	Q_n	$\bar{Q}_n$
1	0	1	0	1
1	1	0	1	0
1	1	1	$\bar{Q}_n$	Q_n

Race around condition

⇒ When both inputs are 1 so output will be Compliment.

- Present State Next State Table

Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

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- Characteristic Eqn

Q_n \ JK	00	01	11	10
0			1	1
1	1			1

$$Q_{n+1} = \overline{Q_n} J + Q_n \overline{K}$$

- Excitation Table

Excitation Ips		O/P	
Q_n	Q_{n+1}	J	K
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0

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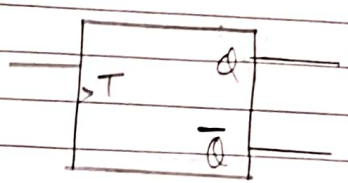
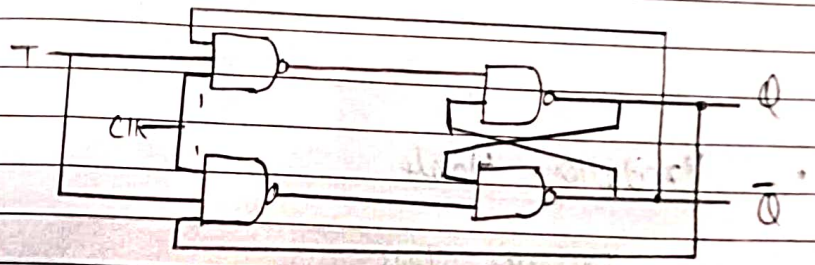
T Flip-Flop [Toggle Flip-Flop]



- Circuit Diagram



- Truth Table

CLK	T	Q_{n+1}	$\overline{Q_{n+1}}$
0	X	Q_n	$\overline{Q_n}$
1	0	Q_n	$\overline{Q_n}$
1	1	$\overline{Q_n}$	Q_n

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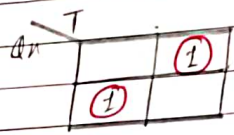
- Present State Next State Table

Qn	T	Qn+1
0	0	0
0	1	1
1	0	1
1	1	0

- Characteristic Eqn



$Q_{n+1} = \overline{Q_n}\,T + Q_n\,\overline{T}$

$\Rightarrow (Q_n \oplus T)$

- Excitation Table

Excitation Tpps		
Qn	Qn+1	T
0	0	0
0	1	1
1	0	1
1	1	0

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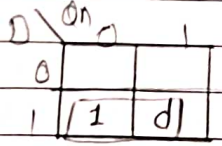
Realization of one flip flop from another (Conversion of flip flop)

① D FF using SR FF

Truth Table & Characteristic Table

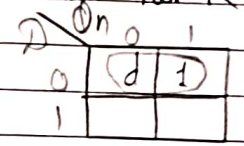
D	Qn	Qn+1	S	R
0	0	0	0	d
0	1	0	0	1
1	0	1	1	0
1	1	1	d	0

- K-MAP for S



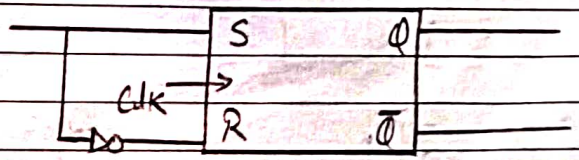
$\Rightarrow S = D$

- K-MAP for R



$\Rightarrow R = \overline{D}$

Circuit Diagram

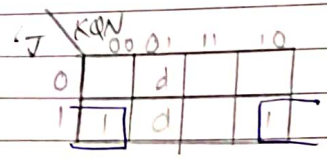


JK ff using SR FF

Truth Table & Characteristic Table (K-MAP)

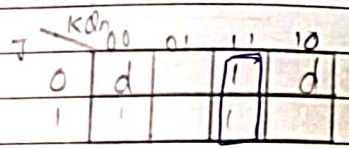
J	K	Qn	Qn+1	S	R
0	0	0	0	0	d
0	0	1	1	d	0
0	1	0	0	0	d
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	d	0
1	1	0	1	1	0
1	1	1	0	0	1

- K-MAP for S



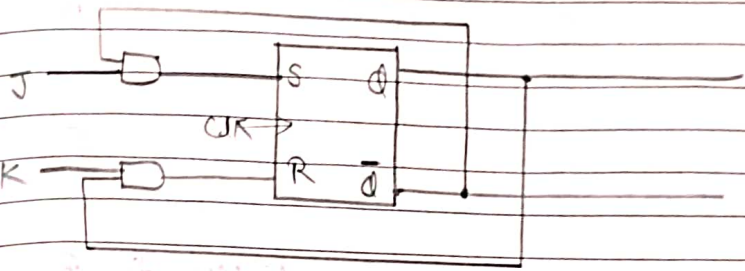
S ⇒ J $\overline{Q_n}$

- K-MAP for R



R = K Qn

Circuit Diagram



(Next State Decoder)

T ff using SR ff :

Characteristic Table

T	Qn	Qn+1
0	0	0
1	0	1
0	1	1
1	1	0

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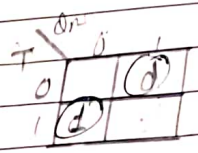
Characteristic Table

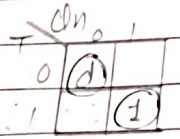
T	Q_n	Q_{n+1}	S	R
0	0	0	0	d
0	0	1	1	0
1	0	1	d	0
0	1	1	0	1
1	1	0	0	

K-MAP for S



$S \Rightarrow T\overline{Q_n}$

K-MAP for R



$R \Rightarrow TQ_n$

Circuit Diagram

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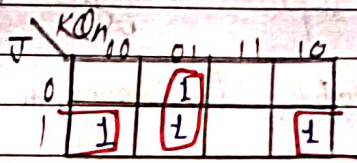
JK ff using D ff

Characteristic Table

J	K	Q_n	Q_{n+1}	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

Output will be 'D'

K-MAP for D



$D \Rightarrow J\overline{Q_n} + \overline{K}Q_n$

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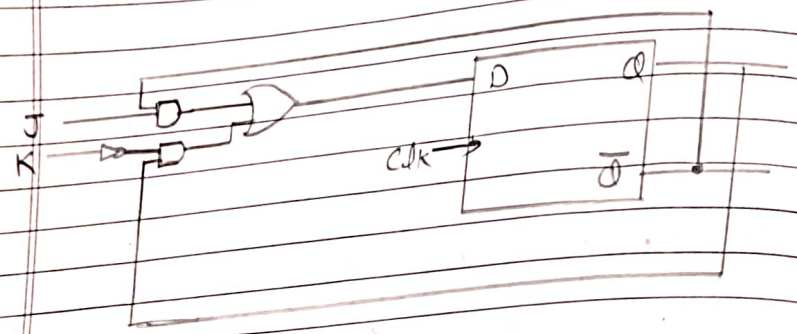
- Circuit Diagram



★ T ff using D ff

D Flip-Flop : When the clock rise from 0 to 1, the value remembered by the flip-flop becomes the value of the D input (Data) at that instant.

T flip-flop : When the clock rises from 0 to 1, the value remembered by the flip-flop either toggles or remains the same depending on whether the T input (Toggles) is 1 or 0.

Characteristic table (Conversion table)

Q_n	T	Q_{n+1}	D
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

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- K-MAP for D :

T \ Q_N	$\overline{Q}_N$	Q_N
$\overline{T}$	0	1
T	1	0

Expression for D would be

$$D = \overline{T} Q_N + T \overline{Q}_N$$

- Circuit Diagram.